

The Indian DeepTech Ecosystem

What to back, how these companies work, where the money comes from and where it exits

Every statistic below is cited to a named source or marked as a data gap — nothing here is estimated without a citation.

Coverage	47 sectors (45 core + 2 emergent) · 13 cities profiled
Evidence base	1,400 companies · 516 investors · 328 incubators · 134 universities
Sources	1,178 named publications and filings
Case evidence	100 company case studies underpin the sector calls

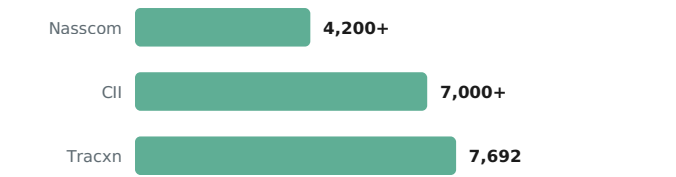
The Answer, in Brief

The ten facts that anchor everything after this page.

#	Finding	What it says
0	The priority call	All 47 sectors scored on two axes. 14 investable today (Back Now), 13 need public capital first (State Must Build), 13 can be left to the market, 7 are neither urgent nor ready (Watch).
1	Funding is re-accelerating	Deep-tech funding hit \$2.3B in 2025, +37% YoY; deep tech's share of all PE/VC quadrupled 4% → 5% (2016–25).
2	AI is a near-monoculture	AI took 84% of deep-tech startups and 91% of deep-tech funding in 2025 — a strength and a single-sector risk.
3	Ecosystem size is a scope question	Nasscom 4,200+ (narrowest); CII 7,000+ (broad); Tracxn 7,692 (broadest). All verified; different inclusion criteria, not a discrepancy.
4	A structural mid-stage bottleneck starves the middle	Seed is accessible but Series A–C lags; only 364 of 1,780 funded companies reached Series A+, 203 Series B+.
5	The capital-scale gap with peers is an order of magnitude	2016–23: India drew \$3.4B of deep-tech VC vs China's \$37.6B (~11×) and the US's \$177.4B (~52×).
6	R&D intensity is the binding constraint	India spends 0.64% of GDP on R&D — roughly a quarter of China's ~2.4–2.7% — with only ~36% privately funded.
7	Patents surged, but grants and monetisation lag	110,375 filings FY2024-25, 61.79% domestic share; grants fell to 33,504; cross-border IP receipts just \$1.53B (vs US \$124B).
8	Long-horizon ambition is now quantified	Viksit Bharat @2047: \$30T economy, 100 GW nuclear, 15% of global space economy; NITI Aayog: top-3 quantum economy by 2035.

Ecosystem size — why the count depends on who's counting, and how India ranks

Same country, three counts — because scope differs



Nasscom, CII, Tracxn — different inclusion criteria, not a discrepancy.

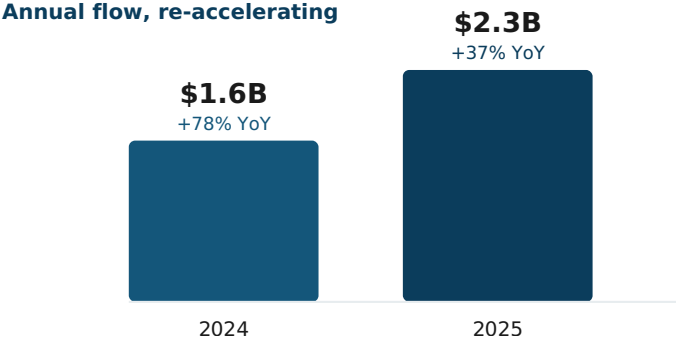
On the broadest count, India ranks 2nd globally



Source: Tracxn (global rank and broadest India count).

The money picture

Annual flow, re-accelerating



Source: Nasscom-Zinnov (annual); separate cumulative/share trackers cited alongside — not spliced into one series.



From funded to scaled — where the pipeline actually breaks



All-time exits on record

● 103 acquisitions ● 40 IPOs — against 1,780 funded companies (Tracxn), all-time

Tracxn funnel, all-time. Underwrites Finding 4: capital thickens at seed, thins sharply by Series A, and the exit ledger stays short.

The Scorecard: Prioritizing 47 Sectors

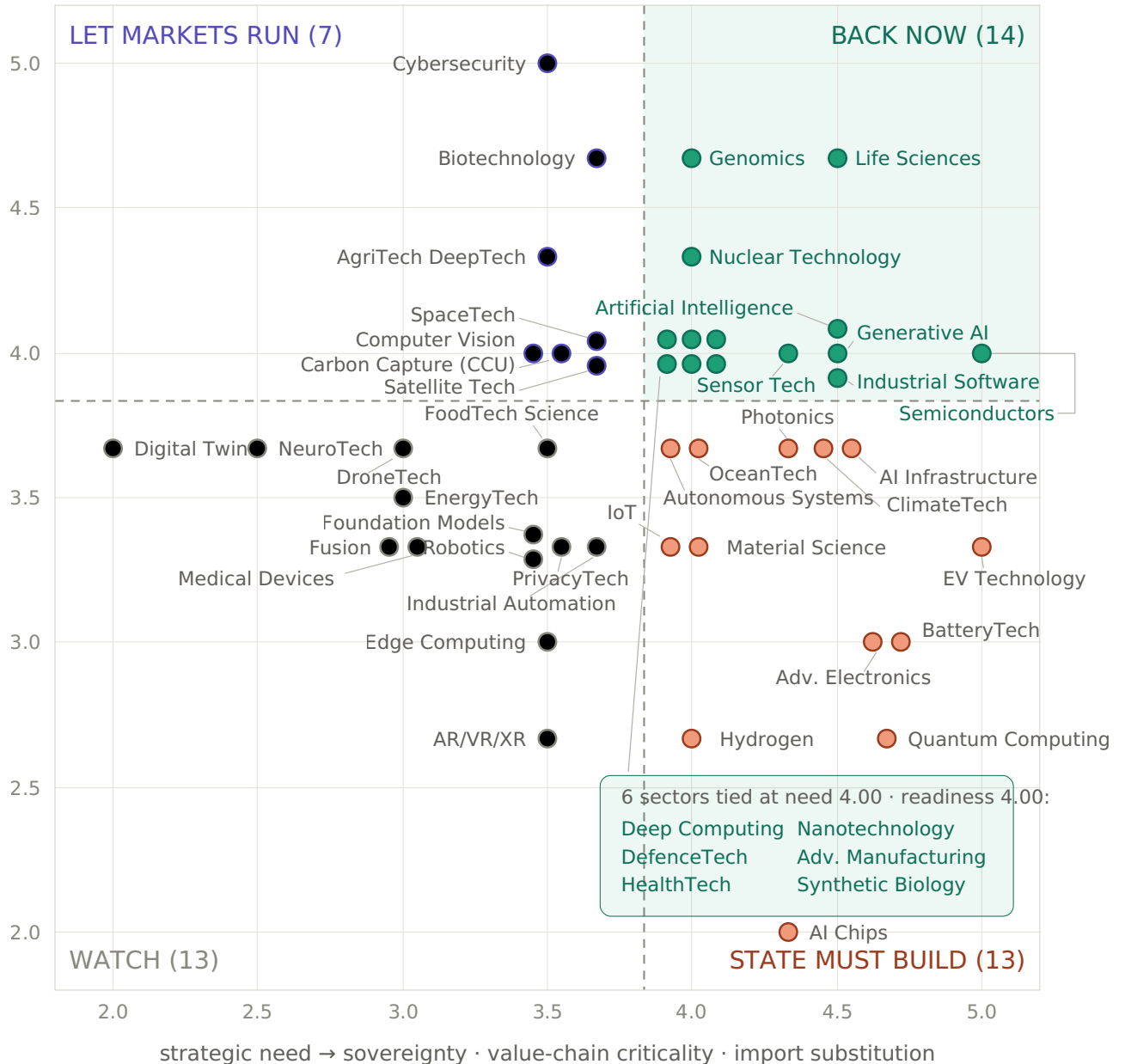
Every sector scored on a 9-criteria rubric, grouped into three axes — Strategic Need, Readiness to Execute, Opportunity — and placed into four quadrants. Composites exclude data-gap criteria from the average rather than imputing them; sectors scored on fewer than 7 of 9 criteria carry no composite ("unranked").

The nine criteria, scored 1 (weakest) to 5 (strongest)

#	Criterion	Axis	What it measures	Scoring bands, 5 → 1
1	Global market size	Opportunity	Worldwide total addressable market for the sector	5 ≥\$500B · 4 \$100–500B · 3 \$20–100B · 2 \$5–20B · 1 <\$5B
2	Domestic market size	Opportunity	India's addressable market for the sector	5 ≥\$50B · 4 \$10–50B · 3 \$2–10B · 2 \$500M–2B · 1 <\$500M
3	Talent availability	Readiness	Depth of India's specialist engineering/PhD pool in this field	By headcount or institutions — 5 ≥50,000 specialists / ≥15 institutions · 4 10k–50k / 8–14 · 3 2k–10k / 4–7 · 2 200–2,000 / 1–3 · 1 <200 / none documented
4	Infrastructure availability	Readiness	Fabs, labs, testbeds, launch and trial facilities in India	5 ≥5 national-scale facilities · 4 2–4 operational · 3 exactly 1 operational · 2 funded/announced, not yet operational · 1 none on record
5	Sovereignty value	Strategic Need	Strategic necessity for India to own this capability domestically	5 dedicated national mission with outlay · 4 named in policy with budget line · 3 named priority, no funding · 2 state-level policy only · 1 no sovereignty framing found
6	Value-chain criticality	Strategic Need	Whether this is a chokepoint others depend on, or a commodity step	5 global supply ≤3 suppliers/countries · 4 input dependency for ≥3 sectors · 3 input to 1–2 sectors · 2 input with substitutes/many suppliers · 1 downstream/commodity
7	Import substitution	Strategic Need	Value of current imports the sector could displace	5 ≥\$10B/yr imported · 4 \$2–10B/yr · 3 \$500M–2B/yr · 2 \$50–500M/yr · 1 <\$50M/yr or net exporter
8	Cluster dependencies	Readiness	Whether viability needs a co-located supplier ecosystem, and whether India has one	5 viable standalone, no cluster needed · 4 cluster needed and exists · 3 cluster exists with gaps · 2 cluster funded/announced, not operational · 1 needs cluster India lacks
9	Growth trajectory	Opportunity	Forecast CAGR, or the CAGR implied by a published target year	5 ≥30% CAGR · 4 20–30% · 3 12–20% · 2 6–12% · 1 <6%

A sector needs 7 of 9 criteria scored to carry a composite; below that it is reported as "unranked" (insufficient evidence), not scored low. Unscored criteria are excluded from the average, never imputed. This is what every "N of 9" and every heatmap column below refers to.

↑ readiness to execute — talent · infrastructure · cluster



All 47 sectors plotted; tied scores carry small offsets. Quadrants are unranked sets.

Axis cutoff ≥ 4.00 of 5 (equivalently 3.67; none fall between). Source: master sector-scoring rubric, v1.5.

Need (x-axis) vs Readiness (y-axis), all 47 sectors labelled. Filled markers = Back Now (teal) and State Must Build (coral); outline markers = Let Markets Run and Watch. Quadrant counts and boundaries are labelled directly on the chart above.

Back Now — 14

High need, India can execute now.

State Must Build — 13

High need, public capital must go first.

Let Markets Run — 7

Real capability, lower strategic necessity.

Watch — 13

Neither urgent nor ready on current evidence.

Sector-by-Sector: What to Back, and What Needs Public Capital First

Each dossier: composite score and criteria coverage, Need/Readiness/Opportunity (1–5), the dominant business model, the sector's own opening finding, and all four capital-pathway rungs with a named, real transaction at each — entry, bridge to Series A, growth capital, and the realistic exit (dashed dot = gap or not yet completed).

Back Now — investable today (14 sectors)

Heatmap — all 14 Back Now sectors, nine criteria



Composite 4.33 · 9 of 9 criteria · 41 companies tracked



Business model: Fabless chip design + IP, outsourced fabrication

Three criteria drive the placement, and all three are 5s. Sovereignty value fires the top rule — its own national mission with an allocated outlay — on the India Semiconductor Mission's ₹76,000 crore incentive framework under MeitY [30].

SCORE BREAKDOWN — ALL 9 CRITERIA

STRATEGIC NEED	READINESS	OPPORTUNITY
● Sovereignty 5/5: dedicated national mission with outlay ● Value-chain 5/5: global supply ≤3 suppliers/ countries ● Import substitution 5/5: ≥\$10B/yr imported	● Talent 5/5: ≥50,000 specialists / ≥15 institutions ● Infrastructure 4/5: 2–4 operational ● Cluster deps. 3/5: cluster exists with documented gaps	● Global market 5/5: ≥\$500B ● Domestic market 4/5: \$10–50B ● Growth trajectory 3/5: 12–20% CAGR

INFRASTRUCTURE & CLUSTER

Present: three ATMP/OSAT units in commercial production, all at Sanand [35]. Absent: an operating wafer fab — Dholera is still construction-stage [35] — and, more bindingly, ultra-high-purity chemistry.

CAPITAL PATHWAY



Common entry point: a government scheme or mission, open to any qualifying company in the sector — not a private deal specific to the named company below.

● **Entry:** DLI Scheme — 50% of eligible expenditure, capped ₹15 crore per application, for fabless design startups

No completed bridge-stage transaction is documented for this sector.

● **Bridge:** [DATA GAP — needs round-level disclosure for Indian fabless seed – A transitions]

No completed growth-stage transaction is documented for this sector.

● **Growth:** [DATA GAP — needs Series B+ round evidence]

Not enough evidence to call this a repeatable pattern — this is the only completed exit on record for the sector (India has completed exits in just 30 of 47 tracked sectors).

● **Exit:** NXP Semiconductors acquired Kinara, Indian-operated fabless edge-AI designer, for \$307M cash

WHO IS BUILDING

Mindgrove Technologies — Chennai · Series A · \$8.0M	AGNIT Semiconductors — Bengaluru · \$4.9M	Maieutic — Seed · \$4.2M
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SHIPS: **Mindgrove Technologies** — Secure IoT is India's first commercial high-performance SoC, an IIT Madras Shakti RISC-V-based microcontroller for · **AGNIT Semiconductors** — GaN-on-Si power epiwafers and transistors for EV fast-chargers and data-center power supplies, sold to · **Maieutic** — A GenAI-powered copilot that automates analog IC design workflows, sold to chip-design teams to speed up analog

TALENT SOURCING — COMMON PROPERTY BIMODAL-FOUNDER-PATHS

Across 9 companies researched, founder background does not converge on one pipeline — it splits roughly in two. One cluster is pure academic-lab spinout (Mindgrove from IIT Madras's Shakti RISC-V project; AGNIT from 18 years of IISc GaN research; Morphing Machines from IISc professor S.K. Nandy and Dr. Ranjani Narayan's decades of computer-architecture research), where the lab work became the company. The other cluster is multinational-chipmaker-veteran or serial-founder: Tata Electronics' CEO Randhir Thakur spent 5 years as President of Intel Foundry Services (plus stints at Applied Materials, SanDisk, Micron) before joining Tata; Aura Semiconductor's Srinath Sridharan worked as a design manager at Silicon Labs, NXP, and St-Ericsson before founding in 2011; Maieutic's founders are serial entrepreneurs who built and sold Steradian Semiconductors to Renesas in 2022 before starting Maieutic. So the honest finding is two coexisting tracks rather than one dominant thread. On engineering (non-founder) talent, the primary signal is that captive R&D centres for Texas Instruments (since 1985) and Intel (since 1988) feed most design-engineering alumni into these startups, alongside IISc/IIT Madras VLSI programs feeding the academic-spinout cluster specifically.

WHAT TO DO

Investor: underwrite fabless design at the DLI-matched stage — DLI covers 50% up to ₹15 crore [42], so equity is buying the uncovered tape-out and go-to-market spend, not the whole R&D bill.

Benchmark exit to NXP–Kinara at \$307M [43]. Do not model a Series A/B path off precedent: both rungs are [DATA GAP].

Carve a localisation line for ultra-high-purity chemicals and ultrapure water [38] — with three ATMP/OSAT units already producing and the first fab still under construction [35], input readiness, not incentive quantum,...

Composite 3.78 · 9 of 9 criteria · 13 companies tracked

Need  4.3
Readiness  4.0
Opportunity  3.0

Business model: Data-as-a-service from sensor/EO hardware

The placement is an import-substitution and talent case, not a market-size case. Import substitution scores 5 on near-total import dependence: over 90% of the domestic sensor market is imported per ELCIA/IESA [44], with no domestic sensor fab on record [45].

SCORE BREAKDOWN — ALL 9 CRITERIA

STRATEGIC NEED

- **Sovereignty 4/5:** named in policy with budget line
- **Value-chain 4/5:** input dependency for ≥3 sectors
- **Import substitution 5/5:** ≥\$10B/yr imported

READINESS

- **Talent 5/5:** ≥50,000 specialists / ≥15 institutions
- **Infrastructure 4/5:** 2–4 operational
- **Cluster deps. 3/5:** cluster exists with documented gaps

OPPORTUNITY

- **Global market 4/5:** \$100–500B
- **Domestic market 3/5:** \$2–10B
- **Growth trajectory 2/5:** 6–12% CAGR

INFRASTRUCTURE & CLUSTER

Present: SCL Chandigarh's 6-inch MEMS device fabrication line and CeNSE IISc's National Nanofabrication Centre, which has produced moisture, neutron and gyroscope sensors [[49], [48]]. Cluster: South India (Bengaluru/Karnataka), 30% of the India MEMS market in 2025, anchored on the semiconductor design ecosystem [52].

CAPITAL PATHWAY



Common entry point: a government scheme or mission, open to any qualifying company in the sector — not a private deal specific to the named company below.

● **Entry:** NIDHI Seed Support Programme (DST) via recognised TBIs — up to ₹100 lakh, typically ₹10–25 lakh

No completed bridge-stage transaction is documented for this sector.

● **Bridge:** [DATA GAP — needs seed → A round evidence for Indian MEMS/sensor hardware]

Not enough evidence to establish a common investor pattern for growth capital here — this is the only named transaction on record.

● **Growth:** Tonbo Imaging: ₹175 crore (~\$20.4M) Series D from Florintree Advisors, Tenacity Ventures, Export-Import Bank of India

No common exit pattern: the sector's own evidence explicitly documents that no completed acquisition or listing exists yet, consistent with the ecosystem-wide finding that 0 of 516 investors profiled report exit data.

● **Exit:** Tonbo Imaging preparing to file for IPO — stated intention, not a completed listing

WHO IS BUILDING

13 tracked companies, of which one is recorded acquired and one deadpooled — the only sector in this group with both outcomes on the record. HQ is unrecorded for 8 of 13: [DATA GAP — needs company registry/website confirmation].

SHIPS: **GalaxEye** — Mission Drishti, India's largest privately built satellite carrying a fused SAR-optical sensor payload · **Dozee** — Dozee Pro, an under-mattress contactless ballistocardiography sensor sheet that turns any hospital bed into a · **Steradian Semiconductors** — A 76-81 GHz CMOS 4D imaging radar transceiver chipset for automotive ADAS object recognition, acquired by Renesas

TALENT SOURCING — COMMON PROPERTY TWO-FEEDER-NOT-ONE

The single-feeder-pipeline story does not generalize across the sector: it recurs in a corporate-lab form rather than a student-team form. Steradian Semiconductors' founders all came directly from RFIC design roles at Texas Instruments and Qualcomm (15-20+ years before spinning out together in 2016), and Tonbo Imaging's founders (Arvind Lakshmikumar, a CMU Robotics PhD who had headed defense-imaging firm Sarnoff Technologies, plus Sarnoff colleagues) spun out of that single employer in 2012 — both mirror GalaxEye's 'one team walks out together' shape, just with a company instead of a college team as the feeder. Dozee (IIT grads who left simulation firm Altair), Ultrahuman (serial entrepreneurs pivoting from their logistics startup after its sale to Zomato, no sensor-specific background), and Airveda (an ex-Microsoft/Facebook/Zomato product exec teamed with her brother's 15 years of electronics-manufacturing experience) show no shared single feeder at all — each is an individual professional pivot from an unrelated domain. So across 6 companies the honest finding is a split: three show a tight single-feeder-team pattern (student team or corporate R&D team departing en masse), while three show unrelated individual career paths converging on a sensor product; on engineering talent, the clearest signal is that sensor engineering capability in India is built through joint academic-DRDO programmes, e.g. IIT Madras professors and DRDO scientists jointly developing the piezo-MEMS acoustic sensor fabrication process at the DRDO Industry Academia-Ramanujan Centre of Excellence.

WHAT TO DO

Investor: the unoccupied slot is the bridge rung. Entry capital is ₹10–25 lakh NIDHI-SSP tickets [56] and the only growth comp is a ₹175 crore Series D [57] — nothing on record spans them.

A ₹5–25 crore MEMS-hardware bridge vehicle underwrites against a >90% import share [44], the most bankable substitution thesis in this group.

Note that the sector's headline exit is an intention [57]; the only completed transaction is domestic consolidation [58].

Composite 3.50 · 8 of 9 criteria · 3 companies tracked

Need  **4.0**
Readiness  **4.0**
Opportunity  **2.7**

Business model: Domestic HPC/supercomputer hardware manufacturing

This is an infrastructure-led placement. Infrastructure availability scores the maximum 5: 37 national-scale supercomputer systems (PARAM Rudra series and predecessors) run under the National Supercomputing Mission. Sovereignty scores 4 on a dedicated ₹4,500 crore (~\$542M) mission run by MeitY/DST via C-DAC and IISc.

SCORE BREAKDOWN — ALL 9 CRITERIA

STRATEGIC NEED

- **Sovereignty 4/5:** named in policy with budget line
- **Value-chain 4/5:** input dependency for ≥3 sectors
- **Import substitution:** data gap

READINESS

- **Talent 4/5:** 10k–50k specialists / 8–14 institutions
- **Infrastructure 5/5:** ≥5 national-scale facilities
- **Cluster deps. 3/5:** cluster exists with documented gaps

OPPORTUNITY

- **Global market 3/5:** \$20–100B
- **Domestic market 3/5:** \$2–10B
- **Growth trajectory 2/5:** 6–12% CAGR

INFRASTRUCTURE & CLUSTER

Present, and at scale: 37 operational NSM systems [59], plus AIRAWAT at C-DAC Pune (200 AI petaflops standalone, 410 combined with PARAM Siddhi-AI) [[59], LOW confidence].

CAPITAL PATHWAY



No dilutive entry rung at all — this sector's companies bypass external capital entirely at this stage.

● **Entry:** No dilutive rung — Netweb raised no external VC/PE round before IPO; entry and growth funded by government/enterprise procurement revenue

Not enough evidence to establish a common investor pattern for bridge capital here — this is the only named transaction on record.

● **Bridge:** No VC-backed bridge on record — the sector bypasses dilutive early-stage capital

Not enough evidence to establish a common investor pattern for growth capital here — this is the only named transaction on record.

● **Growth:** No growth-stage VC/PE on record: zero external rounds straight to public markets

Not enough evidence to call this a repeatable pattern — this is the only completed exit on record for the sector (India has completed exits in just 30 of 47 tracked sectors).

● **Exit:** Netweb Technologies listed on NSE and BSE, 27 July 2023, ₹631 crore IPO

WHO IS BUILDING

Three tracked companies is the thinnest private base of the four sectors — a real finding, not only a coverage gap: the sector's capability sits in public institutions (C-DAC, IISc, IITs) and one listed private champion (Netweb Technologies), not a startup cohort.

SHIPS: **Netweb Technologies** — The 'Tyrone Camarero' server range, with over 200 dual-processor server models, sold as part of India's largest · **QpiAI** — QpiAI-Indus, India's first full-stack quantum computer — a 25-qubit superconducting system deployed under the · **AlphaICs** — Gluon, the first chip built on AlphaICs' Real AI Processor (RAP) architecture, delivering 8 TOPS of edge-AI

TALENT SOURCING — COMMON PROPERTY PATTERN-SPLITS-ON-ORIGIN-NOT-SECTOR

The 'veteran operators, not researchers' pattern holds for the enterprise-hardware/quantum-systems cluster: Netweb's Sanjay Lodha (running his own hardware business since 1996), QpiAI's and AlphaICs' shared founder Nagendra Nagaraja (25 years across NVIDIA, Qualcomm and HMicro), and AlphaICs co-founder Vinod Dham (Intel's 'father of the Pentium') are all decades-long industry veterans, not researchers. But it breaks once academic spinouts are added: InCore Semiconductors grew directly out of the IIT Madras Shakti microprocessor research project, with co-founders coming from academia rather than corporate careers, and BosonQ Psi's founders went straight from HPC-simulation graduate research (Rensselaer, Minnesota) into founding the company with no prior industry tenure. So the pattern is real but conditional on company origin — hardware-manufacturing and chip/quantum-systems ventures selling into enterprise/government skew veteran-operator, while academic-lab spinouts skew the opposite way — rather than a sector-wide rule. On engineering talent, IIT Madras's Shakti project directly feeds hands-on RISC-V/chip engineers into InCore Semiconductors and, via incubation, into QNu Labs, lining up with Tracxn's broader finding that HPC/deep-computing founders cluster from BITS Pilani, IIT Delhi and IIT Kanpur.

WHAT TO DO

Investor: accept that this sector does not currently pay venture economics on the record.

Zero pre-IPO rounds, no growth round, one listing and no acquirer on record [[64], [65]] means the investable expressions are pre-IPO/public equity in systems OEMs and picks-and-shovels around the indigenous-processor...

Government: NSM has converted ₹4,500 crore into 37 fielded systems [59] — a delivery record — but the demand it creates still leaks offshore at the processor layer [59].

Composite 3.62 · 8 of 9 criteria · 7 companies tracked

Need  **4.0**
Readiness  **4.0**
Opportunity  **3.0**

Business model: Patent-protected hardware for institutional/govt buyers

Nanotechnology is the only sector in this group whose cluster dependency is fully met (scores 4): the INUP-i2i network of 6 MeitY nanocentres (IISc, IIT Bombay, Delhi, Guwahati, Kharagpur, Madras) already provides shared cleanroom infrastructure, which also fires the maximum infrastructure score of 5.

SCORE BREAKDOWN — ALL 9 CRITERIA

STRATEGIC NEED

- **Sovereignty 4/5:** named in policy with budget line
- **Value-chain 4/5:** input dependency for ≥3 sectors
- **Import substitution:** data gap

READINESS

- **Talent 3/5:** 2k–10k / 4–7 institutions
- **Infrastructure 5/5:** ≥5 national-scale facilities
- **Cluster deps. 4/5:** cluster needed and exists in India

OPPORTUNITY

- **Global market 4/5:** \$100–500B
- **Domestic market 2/5:** \$500M–2B
- **Growth trajectory 3/5:** 12–20% CAGR

INFRASTRUCTURE & CLUSTER

Present: the INUP-i2i network — 6 MeitY-initiated nanocentres at IISc Bangalore, IIT Bombay, IIT Delhi, IIT Guwahati, IIT Kharagpur and IIT Madras, currently operating [66].

CAPITAL PATHWAY



Common entry point: a government scheme or mission, open to any qualifying company in the sector — not a private deal specific to the named company below.

● **Entry:** No dedicated instrument. DST Nano Mission (2007–17) is closed to new projects and converted to a monitoring-only National Programme; entry falls back on general schemes, e.g. DST NIDHI-PRAYAS up to ₹40 lakh

No sector-specific investor pattern evidenced — Inflexor Ventures is a generalist/opportunistic backer here, not a specialist. This is the only named transaction on record for this rung.

● **Bridge:** NoPo Nanotechnologies' Pre-Series A, co-led by Axilor's Micelio Fund and Inflexor Ventures

Not enough evidence to establish a common investor pattern for growth capital here — this is the only named transaction on record.

● **Growth:** Thin but not absent: Tracxn shows 5 of 163 tracked Indian nanotech companies have reached Series B+; 9 reached Series A+; 43 funded in total [75]. Cumulative Series A across tracked companies \$9.33M [[76], LOW confidence]

No common exit pattern: this is one of the majority of sectors with zero completed exits on record. Ecosystem-wide, 0 of 516 investors profiled report any exit data — the single biggest caveat on every route in this report.

● **Exit:** [DATA GAP — needs a completed exit transaction]

WHO IS BUILDING

NoPo Nanotechnologies — Bengaluru · Seed · \$3.0M

Shilps Sciences — Bengaluru · Pre-Series A

NanoSniff Technologies — Mumbai

SHIPS: **NanoSniff Technologies** — NanoSniffer, the world's first microsensor-based explosive trace detector, capable of detecting Nitroglycerine · **InnoNano Research** — AMRIT (Arsenic and Metal Removal by Indian Technology), a nanomaterial-based gravity-fed water filter that removes · **FIBSOL Life Technologies** — N-FIB 20, a 5-gram water-soluble nanofiber sachet that delivers compacted (1000x) soil-beneficial microbes to

TALENT SOURCING — COMMON PROPERTY IIT-FACULTY-FOUNDER-MAJORITY

Adding five more companies to the earlier 2-company sample surfaces a real thread the smaller sample missed: the dominant founder path in Indian nanotech is the inventing academic spinning a company directly out of their own lab. T. Pradeep (IIT Madras chemistry professor) founded both InnoNano Research (2004) and Hydromaterials (2017) straight from his water-nanomaterials research; NanoSniff was co-founded by IIT Bombay professors alongside Dr. Nitin Kale, who moved from a TSMC process-engineering role (32nm High-K Metal Gate, 2007-08) into managing IIT Bombay's nanofabrication facility before co-founding in 2011; QnanoSol grew out of a DNA-nanotechnology faculty lab at IIT Gandhinagar; and FIBSOL was founded by two IIT Madras PhD scholars straight out of their doctoral research. That leaves Shilps Sciences (PhD EPFL + 20 years in Swiss/Canadian instrumentation research before founding) and NoPo Nanotechnologies (self-driven from a space-enthusiast hobby with no prior corporate employer) as genuine outliers rather than the norm: across all seven companies, 5 of 7 founders came directly from an IIT lab or PhD, making 'inventing-faculty-as-founder' the closest thing to a real pattern. Engineering (non-founder) talent is consistently organized as a cross-disciplinary team spanning sensor design/nano-fabrication, surface functionalization, microfluidics, and embedded hardware-software co-design (illustrated by NanoSniff's team structure), with IIT-incubated PhD scholars and postdocs supplying that talent elsewhere in the sector too.

WHAT TO DO

Investor: this is a capex-light entry in disguise — INUP-i2i's 6 centres absorb the fabrication capex [66], so pre-Series A cheques buy application development rather than cleanrooms, and a deep-tech syndicate precedent...

Government: two distinct fixes. (i) Restore a dedicated entry instrument — the Nano Mission is closed to new projects and the fallback is a ₹40 lakh NIDHI-PRAYAS ticket [73], which does not fund a nanofabrication...

Corporate: be the missing exit.

Composite 3.75 · 8 of 9 criteria · 21 companies tracked

Need  **4.0**
Readiness  **4.7**
Opportunity  **2.7**

Business model: B2B testing & bioinformatics services

Genomics scores 3.75 composite (8 of 9 criteria). The chokepoint case is strong — Illumina holds ~80% of global DNA sequencing — but the domestic market is still small (\$500M–\$2B) and import substitution is a data gap.

SCORE BREAKDOWN — ALL 9 CRITERIA

STRATEGIC NEED

- **Sovereignty 3/5:** named priority, no dedicated funding
- **Value-chain 5/5:** global supply ≤3 suppliers/ countries
- **Import substitution:** data gap

READINESS

- **Talent 5/5:** ≥50,000 specialists / ≥15 institutions
- **Infrastructure 5/5:** ≥5 national-scale facilities
- **Cluster deps. 4/5:** cluster needed and exists in India

OPPORTUNITY

- **Global market 3/5:** \$20–100B
- **Domestic market 2/5:** \$500M–2B
- **Growth trajectory 3/5:** 12–20% CAGR

INFRASTRUCTURE & CLUSTER

Cluster dependencies = 4 — the dependency met by existing Indian cluster token. Genome Valley Hyderabad is India's premier biotech cluster and already serves genomics-adjacent vaccine, biologics and R&D activity for domestic and multinational firms [129];

CAPITAL PATHWAY



Common entry point: a government scheme or mission, open to any qualifying company in the sector — not a private deal specific to the named company below.

● **Entry:** BIRAC Biotechnology Ignition Grant (BIG) — up to ₹50 lakh grant-in-aid, 18 months, to proof-of-concept [108]

Not enough evidence to establish a common investor pattern for bridge capital here — this is the only named transaction on record.

● **Bridge:** C-CAMP's BIRAC-funded bio-incubator — 450 startups funded, mentored or incubated [115]

Not enough evidence to establish a common investor pattern for growth capital here — this is the only named transaction on record.

● **Growth:** International life-science investors — Novo Holdings led a \$50M investment in MedGenome, South Asia's leading genetic diagnostics and data company [128]

Not enough evidence to call this a repeatable pattern — this is the only completed exit on record for the sector (India has completed exits in just 30 of 47 tracked sectors).

● **Exit:** Strategic acquisition — Reliance Strategic Business Ventures acquired a majority stake in Strand Life Sciences, ₹393 crore, 2021 [109]

WHO IS BUILDING

HaystackAnalytics — Mumbai · \$16.0M

MedGenome — Bengaluru

Mapmygenome — Hyderabad

SHIPS: **MedGenome** — Offers a large portfolio of named, in-house-developed genetic tests spanning rare/inherited disease, oncology and · **Mapmygenome** — Genomepatri, a direct-to-consumer predictive whole-genome test reporting lifestyle, disease-risk and · **HaystackAnalytics** — Omega Suite genomic-analysis platform, including OmegaTB, an NGS-based tuberculosis drug-susceptibility test sold

TALENT SOURCING — COMMON PROPERTY INDUSTRY-NOT-ACADEMIA

Strand Life Sciences' all-sitting-faculty founding team looks like the exception, not the rule for the sector: across the eight newly researched genomics/diagnostics companies, founders overwhelmingly arrive from industry and entrepreneurial backgrounds rather than academia. MedGenome's Sam Santhosh (a repeat founder of SciGenom and AgriGenome Labs) and Mapmygenome's Anu Acharya (13 years running her own genomics-outsourcing firm, Ocimum Bio Solutions, before founding Mapmygenome) are serial entrepreneurs; 4baseCare's Hitesh Goswami spent 20 years in pharma drug discovery at Piramal Life Sciences before co-founding Bionivid and then 4baseCare, and LifeCell was built by a pharma-industry family moving into cord-blood banking, not by lab researchers. Even Genotypic's Raja Mugasimangalam, despite a genomics postdoc pedigree (Weizmann Institute, Argonne National Lab), only founded his company after a stint at an Israeli genomics company (QBI) rather than straight from a university post; the one clear sitting-faculty co-founder found, IIT Bombay's Prof Kiran Kondabagil at HaystackAnalytics, sits alongside two non-academic co-founders, so even that case is not a pure academic spinout. Engineering/bioinformatics talent sourcing turned up no evidence beyond the fallback: it reportedly originated from an IISc computer-science applications lab (PerCoLat) that pivoted into computational biology, seeding Strand's technical team.

WHAT TO DO

Investor: Underwrite the exit, not the science. Genomics is the rare bio sector with a closed strategic exit and a named domestic acquirer [109] plus international growth capital already active [128].

Government: The binding constraint is outlay, not capability. Genomics scores 5 on talent and 5 on infrastructure but is capped at 3 on sovereignty solely by a ₹20 crore dedicated line [122].

Corporate: The acquirer precedent is set by an Indian conglomerate buying a genomic-testing and bioinformatics asset [109]. Corporates dependent on the multinational reagent/instrument stack [118] should treat domestic sequencing and bioinformatics capacity as...

Composite 3.38 · 8 of 9 criteria · 8 companies tracked

Need 4.0
Readiness 4.0
Opportunity 2.3

Business model: Direct sale of biomanufactured ingredients

Synthetic Biology scores 3.38 composite (8 of 9 criteria), carried by state-built capacity rather than market pull. As of 2023 the sector had not yet found a mainstream commercial application at scale — infrastructure and mission backing are ahead of the market.

SCORE BREAKDOWN — ALL 9 CRITERIA

STRATEGIC NEED

- **Sovereignty 4/5:** named in policy with budget line
- **Value-chain 4/5:** input dependency for ≥3 sectors
- **Import substitution:** data gap

READINESS

- **Talent 3/5:** 2k–10k / 4–7 institutions
- **Infrastructure 5/5:** ≥5 national-scale facilities
- **Cluster deps. 4/5:** cluster needed and exists in India

OPPORTUNITY

- **Global market 2/5:** \$5–20B
- **Domestic market 2/5:** \$500M–2B
- **Growth trajectory 3/5:** 12–20% CAGR

INFRASTRUCTURE & CLUSTER

Cluster dependencies = 4. The NCR Biotech Science Cluster in Faridabad — an established Indian cluster — hosts THSTI and RCB, the institutions anchoring synthetic-biology-relevant biomanufacturing [139], and a BioE3-funded precision biotherapeutics biofoundry (mAbs, live biotherapeutics) sits at THSTI [130].

CAPITAL PATHWAY



Common entry point: a government scheme or mission, open to any qualifying company in the sector — not a private deal specific to the named company below.

● **Entry:** BIRAC BIG — the same instrument as core biotechnology, up to ₹50 lakh for 18 months, accessed via BIG partners such as C-CAMP [108]

Not enough evidence to establish a common investor pattern for bridge capital here — this is the only named transaction on record.

● **Bridge:** C-CAMP's BIRAC-funded bio-incubator (450 startups funded/mentored/incubated), used by synbio ventures including String Bio [115]

Not enough evidence to establish a common investor pattern for growth capital here — this is the only named transaction on record.

● **Growth:** BIRAC-RDI Fund, ₹2,000 crore, funding TRL4+ translational projects including industrial and synthetic biology [138]

No common exit pattern: this is one of the majority of sectors with zero completed exits on record. Ecosystem-wide, 0 of 516 investors profiled report any exit data — the single biggest caveat on every route in this report.

● **Exit:** [DATA GAP — no completed exit transaction found]

WHO IS BUILDING

Sea6 Energy — Bengaluru · Growth · \$27.5M

Fermbox Bio — Bengaluru

Zumutor Biologics — Bengaluru

SHIPS: String Bio — String Pro, a fermentation-derived high-quality protein made from a methane-based feedstock via the company's · **Pandorum Technologies** — Pandorum's 3D-bioprinted human liver tissue construct, engineered from progenitor cells, is supplied to · **Sea6 Energy** — SeaCombine, a mechanized ocean-going rig that simultaneously harvests and replants tropical red seaweed, enabling

TALENT SOURCING — COMMON PROPERTY VETERAN-ANCHOR TO VETERAN-SATURATED

The 'veteran-anchored' pattern holds cleanly for Sea6 Energy: IIT Madras students who began the company as a 2008 iGEM project paired with their professor Shrikumar Suryanarayan, an ex-Biocon veteran who became MD, mirroring String Bio and Pandorum's one-veteran-plus-junior-cofounders shape. But Zumutor Biologics and Jananom show a distinct 'veteran-saturated' variant rather than a single anchor: Zumutor's leadership and advisory bench is stacked with senior operators (CEO plus a Dana-Farber/Harvard/MIT/Bayer advisory network), and Jananom is chaired by a self-described serial synthetic-biology entrepreneur, backed by directors with DARPA/biothreat-program and global precision-fermentation pedigrees. StrainX Bioworks, Sabran Bioentri, Bioaptagen Labs and Mandala Bioworks returned no public founder, product, or funding information and are excluded rather than guessed at. On engineering talent, a signal beyond the aerospace-crossover fallback emerged: Jananom's scale-up/manufacturing lead came directly from manufacturing roles at global precision-fermentation firms Perfect Day, Evolva and Amyris, suggesting India's synbio engineering pipeline, like its founder pipeline, also draws returnee operators from the same US biobased-industry companies.

WHAT TO DO

Investor: Treat the missing exit as the deal term, not a research gap to wish away — no completed exit and no named acquirer are on record for this sector.

Government: The build is done; the demand is not. Six biofoundries are operational [130] while synthetic biology had not yet entered mainstream Indian industrial-chemicals manufacturing as of 2023 [133].

Corporate: India's biofoundries are open capacity in a sector where synbio is a documented input to healthcare, agriculture, environmental science and industrial biotechnology [132].

Composite 4.00 · 8 of 9 criteria · 57 companies tracked

Need  4.0
Readiness  4.0
Opportunity  4.0

Business model: B2B/B2C SaaS and marketplace commissions

HealthTech is the strongest composite in the bio family: 4.00 on 8 of 9 criteria, and the only one of the four balanced across all three lenses — need 4.00 / readiness 4.00 / opportunity 4.00. It is the sole bio sector where the opportunity axis is not the drag.

SCORE BREAKDOWN — ALL 9 CRITERIA

STRATEGIC NEED

- **Sovereignty 4/5:** named in policy with budget line
- **Value-chain 4/5:** input dependency for ≥3 sectors
- **Import substitution:** data gap

READINESS

- **Talent 3/5:** 2k–10k / 4–7 institutions
- **Infrastructure 5/5:** ≥5 national-scale facilities
- **Cluster deps. 4/5:** cluster needed and exists in India

OPPORTUNITY

- **Global market 4/5:** \$100–500B
- **Domestic market 4/5:** \$10–50B
- **Growth trajectory 4/5:** 20–30% CAGR

INFRASTRUCTURE & CLUSTER

Cluster dependencies = 4. C-CAMP Bangalore's Digital Health Platform gives startups access to an existing deeptech cluster — a served-by-existing-cluster dependency structure [146].

CAPITAL PATHWAY



Common entry point: private angel/VC money, not a government scheme — the named round below is a representative instance of a private-capital-first sector.

● **Entry:** Conventional angel/seed VC, not deeptech grant capital — 1mg's first \$6M from Sequoia Capital, Intel Capital and Omidyar Network [110]

No sector-specific investor pattern evidenced — Sequoia is a generalist/opportunistic backer here, not a specialist. This is the only named transaction on record for this rung.

● **Bridge:** Ordinary VC follow-on, not grant-linked — 1mg raised ₹100 crore in Series B from Maverick Capital Ventures with existing investors Sequoia Capital and Omidyar Network [110]

No completed growth-stage transaction is documented for this sector.

● **Growth:** [DATA GAP — no named growth-stage provider on record for this sector]

Not enough evidence to call this a repeatable pattern — this is the only completed exit on record for the sector (India has completed exits in just 30 of 47 tracked sectors).

● **Exit:** Strategic acquisition by an Indian conglomerate's digital arm — Tata Digital Limited acquired a majority stake in 1MG Technologies, announced June 2021 [111]

WHO IS BUILDING

Qure.ai — Mumbai · \$130.0M

Wysa — Bengaluru · Series B · \$20.0M

eka.care — Bengaluru · Series A · \$19.5M

SHIPS: **Qure.ai** — qXR is an FDA-cleared AI chest X-ray solution that detects up to 29 abnormalities (including TB, pneumothorax · **Wysa** — Wysa is an AI-powered mental wellbeing chatbot delivering CBT/DBT-based self-guided support, used by over 6 · **eka.care** — eka.care is an ABHA-integrated personal health record app that lets Indian patients digitize and share medical

TALENT SOURCING — COMMON PROPERTY TWO-TRACK-FOUNDER-PATTERN

The companies split into two distinct founder patterns rather than one universal thread. Software/platform healthtech founders arrive after a substantial multi-year corporate career: Qure.ai's Prashant Warier was Chief Data Scientist at Fractal Analytics before founding; Wysa's Ramakant Vempati spent years in the Tata Administrative Service, Booz & Co. and Goldman Sachs International, while co-founder Jo Aggarwal was Managing Director of Pearson Learning Solutions; eka.care's Vikalp Sahni was Co-Founder/CTO of Goibibo for eight years after starting as an IBM software engineer; Tricog Health's Dr. Charit Bhograj was a practicing cardiologist at Fortis Hospitals and co-founder Dr. Zainul Charbiwala was an IBM Research scientist in embedded medical-tech before founding. In contrast, the smaller diagnostics ventures (Curex Technologies among them) look more like direct academic-lab spinoffs, with PhD scientists moving from a research post straight into a company built on their own lab's IP, with no intervening corporate career. On engineering (non-founder) talent, no healthtech-specific institute or alumni pipeline beyond the documented 1mg/IIT Delhi data point surfaced, so that remains the only sourced fact rather than a broader pipeline claim.

WHAT TO DO

Investor: This is the family's conventional-VC sector: seed and Series B both ran through standard funds [110], the exit is closed and strategic [111], and the model is SaaS/marketplace [148]. Underwrite against India's 25.12% CAGR [141].

Government: ABDM's ₹1,600 crore [143] built six national platforms [142] — the highest-leverage next move is widening third-party access to those rails, since the cluster evidence shows startups already consuming platform access rather than physical infrastructure [146].

Corporate: Tata Digital's acquisition of 1MG [111] is the template: conglomerates with consumer distribution buying digital-health platforms. Corporates should also read the input evidence — digital health is a documented input into medical devices and at least two...

Composite unranked · 6 of 9 criteria · 34 companies tracked

Need  4.5
Readiness  4.7
Opportunity  2.0

Business model: Fee-for-service CRDMO for global pharma clients

Life Sciences shows the highest readiness of the four bio sectors (5.00) but carries no composite: 4 of 9 criteria are data gaps. Every surviving score is a 4 or 5, though three carry a written caveat about scope or double-counting.

SCORE BREAKDOWN — ALL 9 CRITERIA

STRATEGIC NEED

- **Sovereignty 5/5:** dedicated national mission with outlay
- **Value-chain 4/5:** input dependency for ≥3 sectors
- **Import substitution:** data gap

READINESS

- **Talent 5/5:** ≥50,000 specialists / ≥15 institutions
- **Infrastructure 5/5:** ≥5 national-scale facilities
- **Cluster deps. 4/5:** cluster needed and exists in India

OPPORTUNITY

- **Global market:** data gap
- **Domestic market:** data gap
- **Growth trajectory 2/5:** 6–12% CAGR

INFRASTRUCTURE & CLUSTER

Infrastructure is the strongest verified cell: a biosafety-containment network anchored by NIV Pune, India's only operational civilian BSL-4 laboratory [152], with corroborating facilities [153].

CAPITAL PATHWAY



Common entry point: a government scheme or mission, open to any qualifying company in the sector — not a private deal specific to the named company below.

● **Entry:** National Biopharma Mission — running since 2017 with World Bank co-financing, supporting biotech research and biopharmaceutical development [112]

Not enough evidence to establish a common investor pattern for bridge capital here — this is the only named transaction on record.

● **Bridge:** BIRAC SEED Fund, up to ₹30 lakh equity, bridging promoter capital to angel/VC ahead of Series A [113]

Not enough evidence to establish a common investor pattern for growth capital here — this is the only named transaction on record.

● **Growth:** Dedicated healthcare/life-sciences investors — Eight Roads Ventures, with \$1.6B under management in India across its portfolio [158]

Not enough evidence to call this a repeatable pattern — this is the only completed exit on record for the sector (India has completed exits in just 30 of 47 tracked sectors).

● **Exit:** Public listing, not acquisition — Sai Life Sciences, a CRDMO, completed its IPO [114]

WHO IS BUILDING

Aten Porus Lifesciences —

SHIPS: Aten Porus Lifesciences — Rare-disease drug discovery platform (subsidiaries Oraxion and Straxyn) developing gene-delivery and · **Clearysynth Labs** — Certified pharmaceutical reference standards and stable isotope-labeled (deuterium) compounds, supplied to 5,000+ · **Exocan Healthcare Technologies** — Exosome-based liquid-biopsy oncodiagnostics platform for early, lower-cost cancer detection, aimed at oncologists

TALENT SOURCING — COMMON PROPERTY VETERANS-VS-PHD-SPINOUTS-SPLIT

Founder backgrounds split into two distinct tracks rather than one uniform pattern. Capital-intensive manufacturing/ingredient-supply businesses are led by decades-long pharma-industry veterans: Clearysynth's Vijay Ambati had 30+ years in pharmaceutical/biopharma regulatory and analytical-reference-standards work before founding the company in 2009, and Innovation Biologicals' Subhash Kapre spent a 40-year career at Serum Institute of India (rising to Executive Director) before starting his vaccine-ingredient venture. Diagnostics, sensor, and drug-discovery startups skew the opposite way, toward founders straight out of doctoral work: Exocan's Aman Sharma founded during his PhD at University of Pune, Molqbits' Girish Arabale founded right after his NCL Pune PhD in carbon nanostructures, and Aten Porus co-founder Aditya Kulkarni came straight from a Purdue PhD in chemical biology. The "pharma veteran" thread documented for Sai Life Sciences (John Pavey, PhD Organic Chemistry Liverpool, ex-big-pharma CMC leadership) generalizes to the manufacturing/ingredient-supply side of the sector but not to diagnostics or discovery-stage biotech, where academic PhD spinouts are equally common.

WHAT TO DO

Investor: Do not size this sector from the headline market figures above. Both market cells are struck and the India/global pair is internally contradictory. What is actionable is narrower and well-evidenced: the CRDMO business — fee-for-service to 280+ global innovator clients [157],...

Government: Two separate actions. (a) Resolve the definition. The finding here is that Life Sciences may not be separately measurable from Biotechnology, HealthTech, Medical Devices and Genomics — the GCC talent figure is double-count-flagged [150] and the flagship...

Corporate: The evidenced corporate relationship in this sector is customer, not acquirer — 280+ global innovator pharma/biotech clients buying contract R&D and manufacturing [157], with API capacity expanding at Bidar [157].

Composite 3.88 · 8 of 9 criteria · 39 companies tracked



Business model: B2G contracts with Armed Forces/defence PSUs

Three criteria drive the placement: sovereignty fires on a ₹1,39,000 crore FY2026-27 capital-acquisition earmark; infrastructure scores the maximum on DRDO's 52 national labs and test ranges; talent draws on DRDO's ~30,000 scientific and technical staff.

SCORE BREAKDOWN — ALL 9 CRITERIA

STRATEGIC NEED

- **Sovereignty 4/5:** named in policy with budget line
- **Value-chain 4/5:** input dependency for ≥3 sectors
- **Import substitution:** data gap

READINESS

- **Talent 4/5:** 10k–50k specialists / 8–14 institutions
- **Infrastructure 5/5:** ≥5 national-scale facilities
- **Cluster deps. 3/5:** cluster exists with documented gaps

OPPORTUNITY

- **Global market 4/5:** \$100–500B
- **Domestic market 3/5:** \$2–10B
- **Growth trajectory 4/5:** 20–30% CAGR

INFRASTRUCTURE & CLUSTER

Infrastructure scores 5; the cluster scores 3. Both readings are correct and the gap between them is the operating problem.

CAPITAL PATHWAY



Common entry point: a government scheme or mission, open to any qualifying company in the sector — not a private deal specific to the named company below.

● **Entry:** Equity-free grant

Not enough evidence to establish a common investor pattern for bridge capital here — this is the only named transaction on record.

● **Bridge:** Grant-in-aid

Not enough evidence to establish a common investor pattern for growth capital here — this is the only named transaction on record.

● **Growth:** Non-dilutive procurement revenue

Not enough evidence to call this a repeatable pattern — this is the only completed exit on record for the sector (India has completed exits in just 30 of 47 tracked sectors).

● **Exit:** Strategic M&A

WHO IS BUILDING

40 companies tracked. Stage is recorded for 6; the remaining 34 carry name + category + source but no disclosed stage — a disclosure gap, not an absence of activity.

Raphe mPhibr — Noida · Series B

NewSpace Research and Technologies — Bengaluru · Series B

Big Bang Boom Solutions — Chennai · Series C

SHIPS: **Raphe mPhibr** — Indigenous UAVs including the mR20 logistics drone and a 1.5-ton MALE drone, supplied to the Indian Armed Forces · **NewSpace Research and Technologies** — Sheshnaag-150 and Sheshnag-20 loitering munitions plus the MAPSS (Medium Altitude Persistent Surveillance System) · **Big Bang Boom Solutions** — Counter-drone/anti-UAS defence systems and hybrid personal combat armour, sold to Indian defence forces as part of

TALENT SOURCING — COMMON PROPERTY MIXED-FOUNDER-ARCHETYPES-NOT-ONE-PATTERN

Founder backgrounds across the six companies researched are genuinely mixed, not a single pattern: grad-researcher founders who went straight from MIT/Georgia Tech (Raphe mPhibr) or an IIT Delhi student tech-club (Sharang Shakti) into founding; ex-military founders, including a retired IAF fighter pilot at NewSpace Research and Technologies (Sameer Joshi) and a Merchant Navy captain-turned-entrepreneur at Sagar Defence Engineering (Capt. Nikunj Parashar); and veteran corporate/national-lab R&D founders, exemplified by Tonbo Imaging's Arvind Lakshmikummar, who spent over a decade as a research scientist and technology director at Honeywell and Sarnoff Corporation running DARPA/US Army/DRDO imaging programs before founding the company. Big Bang Boom Solutions adds a fourth flavor still — an academic-entrepreneur paired with a business operator (Dr. Ramaswamy Shivaraman and an IIM Lucknow MBA co-founder). Four distinct founder archetypes (grad-researcher, ex-military, veteran-R&D-industry, academic/business-operator) coexist in this sector, not one shared origin story; on engineering talent, Sharang Shakti's core team traces to IIT Delhi engineering-student circles.

WHAT TO DO

Investor: Underwrite the procurement rung, not the grant rung. Diligence should test order-book conversion under DAP-2020 and the working-capital profile of B2G payment cycles [160][169];

Government: The binding constraint is cluster execution, not scheme design. Closing the ~79% implementation shortfall in the Tamil Nadu corridor converts committed capital into usable capacity [167];

Corporate: Zen Technologies and Tata Advanced Systems have both demonstrated that majority-stake acquisition of a UAV/munitions capability is executable in this market [172][173].

Composite 3.89 · 9 of 9 criteria · 3 companies tracked

Need  4.0
Readiness  4.3
Opportunity  3.3

Business model: Pre-commercial, regulator-gated reactor development

India's nuclear capability is world-class and almost entirely state-owned. Its private startup base is three tracked companies. Both statements are true, and conflating them produces the wrong investment conclusion.

SCORE BREAKDOWN — ALL 9 CRITERIA

STRATEGIC NEED

- **Sovereignty 5/5:** dedicated national mission with outlay
- **Value-chain 5/5:** global supply ≤3 suppliers/countries
- **Import substitution 2/5:** \$50–500M/yr imported

READINESS

- **Talent 4/5:** 10k–50k specialists / 8–14 institutions
- **Infrastructure 5/5:** ≥5 national-scale facilities
- **Cluster deps. 4/5:** cluster needed and exists in India

OPPORTUNITY

- **Global market 3/5:** \$20–100B
- **Domestic market 4/5:** \$10–50B
- **Growth trajectory 3/5:** 12–20% CAGR

INFRASTRUCTURE & CLUSTER

The score of 5 is earned by rows 1–3, which a startup cannot use. The commercially reachable asset is row 4 — the Class I component supply chain — and any private nuclear thesis should be built there or on design IP, not on assumed access to national facilities.

CAPITAL PATHWAY



Common entry point: a government scheme or mission, open to any qualifying company in the sector — not a private deal specific to the named company below.

● **Entry:** Statutory permission, not a cheque

Not enough evidence to establish a common investor pattern for bridge capital here — this is the only named transaction on record.

● **Bridge:** Angel/HNI + strategic consortium

Not enough evidence to establish a common investor pattern for growth capital here — this is the only named transaction on record.

● **Growth:** Mission money

No common exit pattern: this is one of the majority of sectors with zero completed exits on record. Ecosystem-wide, 0 of 516 investors profiled report any exit data — the single biggest caveat on every route in this report.

● **Exit:** [DATA GAP]

WHO IS BUILDING

Jindal Nuclear Power Private Limited — Delhi · Growth

IYNS TechSolutions LLP — Pune · Seed

Pranos Fusion — Bengaluru

SHIPS: **Jindal Nuclear Power Private Limited** — Not an indigenous design — a planned ~18 GW utility-scale nuclear buildout across Indian states using licensed EPR · **IYNS TechSolutions LLP** — SOOKSHMA (SUK-M), India's first indigenous thorium-based molten-salt micro modular reactor, targeting 10 MW output · **Pranos Fusion** — A compact high-field magnetic-confinement (tokamak) platform combining plasma-control software and HTS magnets

TALENT SOURCING — COMMON PROPERTY

CONGLOMERATE-VS-LAB-VETERAN-SPLIT

India's nuclear-tech field is not one founder archetype but two distinct tracks, and this is itself the finding for such a thin sector. Jindal Nuclear Power Private Limited has essentially no founder narrative at all — it is a wholly owned subsidiary the Naveen Jindal Group stood up in 2023 to license large reactor designs from EDF, Westinghouse and NPCIL and build roughly 18 GW of conventional capacity, making it a corporate capital-deployment vehicle rather than a deep-tech startup. By contrast, the small independent ventures are led or advised by people drawn from state and international atomic-research institutions rather than classic entrepreneurs: IYNS TechSolutions' key technical advisor is the former Atomic Energy Commission Chairman and BARC Director, while Pranos Fusion's co-founder Shaurya Kaushal holds a PhD from JNCASR and previously worked at the UK Atomic Energy Authority. Engineering talent below the leadership layer is similarly thin and specialist-sourced — IYNS's engineering leadership comes from severe-accident-management and radiological-safety training rather than a startup-engineering pipeline, underscoring how shallow India's non-DAE nuclear talent pool is. Outside the conglomerate track, this sector has almost no public founder-bio culture in the usual startup sense — it is either PSU/corporate-anchored or academic/national-lab-veteran-anchored.

WHAT TO DO

Investor: Treat this as option-value, not a deployable 2026 allocation. There is no completed exit, no named acquirer [186], and the one bridge-stage round on record saw the institutional fund exit rather than lead [185].

Government: The mission money exists (₹20,000 crore) and the legal permission exists (SHANTI, Dec 2025) [177][175]. The missing instrument is access : a defined route by which a private venture can use IGCAR, BARC or NFC facilities and a published...

Corporate: The natural first movers are the firms already pre-qualified for Class I components — Walchandnagar Industries, L&T, BHEL, MTAR Technologies, Godrej & Boyce — for whom SHANTI converts a supply relationship into a potential ownership...

Composite 3.62 · 8 of 9 criteria · 17 companies tracked

Need  **4.0**
Readiness  **4.0**
Opportunity  **3.0**

Business model: Full-stack tech licensing/deployment to industry

Every market cell in this dossier is scoped to additive manufacturing / 3D printing, because that is the scope of the published evidence available at sector level. This matters for one cell in particular.

SCORE BREAKDOWN — ALL 9 CRITERIA

STRATEGIC NEED

- **Sovereignty 4/5:** named in policy with budget line
- **Value-chain 4/5:** input dependency for ≥3 sectors
- **Import substitution:** data gap

READINESS

- **Talent 3/5:** 2k–10k / 4–7 institutions
- **Infrastructure 5/5:** ≥5 national-scale facilities
- **Cluster deps. 4/5:** cluster needed and exists in India

OPPORTUNITY

- **Global market 3/5:** \$20–100B
- **Domestic market 2/5:** \$500M–2B
- **Growth trajectory 4/5:** 20–30% CAGR

INFRASTRUCTURE & CLUSTER

Unusually for this cluster, the infrastructure is genuinely accessible — SAMARTH centres, NCAM and Common Engineering Facility Centres are shared-use by design [188][189][194], which is the opposite of the nuclear position and a materially better starting point than the 21%-implemented defence corridor.

CAPITAL PATHWAY



Common entry point: a government scheme or mission, open to any qualifying company in the sector — not a private deal specific to the named company below.

- **Entry:** Non-dilutive prototyping grant

No completed bridge-stage transaction is documented for this sector.

- **Bridge:** [DATA GAP]

Not enough evidence to establish a common investor pattern for growth capital here — this is the only named transaction on record.

- **Growth:** Strategic equity from global capital-goods OEMs

No common exit pattern: the sector's own evidence explicitly documents that no completed acquisition or listing exists yet, consistent with the ecosystem-wide finding that 0 of 516 investors profiled report exit data.

- **Exit:** No completed acquisition on record. The closest evidenced transactions are minority strategic equity investments into INTECH Additive Solutions (DMG Mori 2018, DN Solutions 2025) — investments, not exits. The National Manufacturing Mission's ₹100 crore outlay is far too small to function as an exit-stage instrument, and the National Strategy for AM sets adoption targets with no stated M&A mechanism [202]

WHO IS BUILDING

17 companies tracked; stage recorded for 3.

Divide By Zero Technologies — Navi Mumbai · Bootstrapped

Ethereal Machines — Bengaluru · Series B

Fabheads Automation — Chennai · Series A

SHIPS: **Ethereal Machines** — Aura and Nimbus, in-house designed and built 3- and 5-axis CNC machining platforms used to manufacture precision · **Fabheads Automation** — FibrBot series continuous carbon-fiber 3D printers (patented Miniaturised Tow Placement process), sold to · **Divide By Zero Technologies** — Aion 500 MK2 and Aecon large-format industrial FDM 3D printers (patented AFPM technology), sold to engineering and

TALENT SOURCING — COMMON PROPERTY MIXED-FOUNDER-ARCHETYPES

Across five companies, Advanced Manufacturing founder backgrounds are a genuine mix of archetypes, not one dominant pattern: self-taught hobbyists (Ethereal Machines' Kaushik Mudda and Navin Jain built CNC routers as engineering-college students before founding in 2014, and Divide By Zero Technologies' Swapnil Sansare ran a self-taught 'bedroom startup'), elite-institution R&D alumni (Fabheads Automation's Dhinesh Kanagaraj and Abhijeet Rathore met at ISRO working on the GSLV Mk-III and Chandrayaan-II programmes for about three years before spinning out in 2015), industry veterans (Intech Additive Solutions' Sridhar Balaram brought over 30 years of foundry/casting experience), and academic-institutional spinouts (Tvasta Manufacturing Solutions grew out of IIT Madras's construction-technology research ecosystem). No single career path predicts who founds a manufacturing deep-tech company in India; separately, the sector faces a documented shortage of highly skilled precision-manufacturing engineering professionals, constraining the engineering-talent pipeline feeding areas such as aerospace manufacturing.

WHAT TO DO

Investor: Underwrite the growth rate, not the current market: India is a ~\$707M market inside a \$30.6B global one growing at 23.9% [192][187].

Government: Two specific fixes. First, the National Strategy for Additive Manufacturing carries targets — 5% global share, \$1B GDP contribution, 100,000 skilled workers — but no allocated budget ;

Corporate: For Indian capital-goods incumbents, the AM layer is an input to six named Focus Sectors — aerospace, defence, automotive, medical devices, capital goods, electronics [191] — which makes it a make-or-buy decision, not a diversification.

Composite 4.25 · 8 of 9 criteria · 135 companies tracked



Business model: Time-and-material services; AI product revenue nascent

Three criteria carry the placement at their maximum: talent (420,000 AI professionals, second-highest globally), sovereignty and value-chain criticality. The binding constraint is infrastructure — India has compute, but only one national-scale pool of it.

SCORE BREAKDOWN — ALL 9 CRITERIA

STRATEGIC NEED	READINESS	OPPORTUNITY
● Sovereignty 5/5: dedicated national mission with outlay ● Value-chain 4/5: input dependency for ≥3 sectors ● Import substitution: data gap	● Talent 5/5: ≥50,000 specialists / ≥15 institutions ● Infrastructure 3/5: exactly 1 operational facility ● Cluster deps. 4/5: cluster needed and exists in India	● Global market 4/5: \$100–500B ● Domestic market 4/5: \$10–50B ● Growth trajectory 5/5: ≥30% CAGR

INFRASTRUCTURE & CLUSTER

One national-scale facility: the IndiaAI Mission common compute facility, 38,000+ GPUs onboarded , operational, offered to startups and academia at affordable rates [79]. Cluster dependency is met domestically — Bengaluru is documented as the world's second-largest AI talent hub [83].

CAPITAL PATHWAY

●

Entry

●

Bridge

●

Growth

●

Exit

WHO IS BUILDING

135 companies tracked. Funded leaders and case-study cohort:

Fractal Analytics — Mumbai · Public · \$341.4M

Mad Street Den — Chennai · Acquired · \$27.5M

Ai Palette — Series A · \$5.5M

SHIPS: **Fractal Analytics** — Cogentiq, an enterprise agentic AI platform providing pre-built autonomous agents, connectors and low-code tools · **Mad Street Den** — Vue.ai, a computer-vision platform for retailers (Macy's, Tata) that auto-tags product images and powers virtual · **NIRAMAI** — Thermalytix, an AI-based medical device for early-stage breast cancer screening, deployed across 200+ hospitals in

TALENT SOURCING — COMMON PROPERTY

VETERAN-LED (WITH EXCEPTIONS)

Roughly half of the founders spot-checked fit the original decade-plus-at-a-global-tech-company mold: NIRAMAI's Geetha Manjunath spent 17 years as a Principal Research Scientist at HP Labs before a stint as Xerox India's Lab Director, Mad Street Den's Ashwini Asokan led Intel Labs' Mobile Innovation Portfolio for about a decade, and RagaAI's Gaurav Agarwal moved through Texas Instruments and NVIDIA before founding. But the pattern is not universal: Fractal Analytics' Srikanth Velamakanni came from investment banking (ANZ, ICICI), ThirdAI's Anshumali Shrivastava is a tenured Rice University professor who spun the company straight out of his research lab, and Skillveri's founders spent their prior careers at an Indian hardware company (Vortex) rather than a global tech giant. So the tech-veteran thread recurs often enough to be worth naming, but it's a common pathway among several, not a sector-wide rule — education (IIT Delhi, Carnegie Mellon, IISc, Rice) is scattered either way. Engineering (non-founder) talent draws on the same GCC-context pool documented ecosystem-wide (120,000+ AI/ML professionals, 185+ Centres of Excellence), with IIT Bombay/Delhi/Madras and IISc alumni disproportionately represented in research-adjacent roles and Bengaluru/Hyderabad as the two largest hiring hubs.

WHAT TO DO

Investor: Underwrite the Sarvam pathway, not the Fractal one. The evidenced ladder runs grant → \$41M Series A → \$234M strategic-led Series B [87][88]; the evidenced exit is a single 2017 trade sale [89].

Government: The score's only sub-4 is infrastructure, and the rule that produced it is exactly one national facility [79]. The marginal rupee of the ₹10,372 crore outlay [79] buys more by adding a second and third compute pool than by adding programmes.

Corporate: HCLTech's \$150M strategic lead [88] is the template: buy the model layer through a growth round, not an acqui-hire. With Google the only on-record acquirer [89], an Indian corporate entering now sets the price.

Generative AI

BACK NOW

Composite 3.88 · 8 of 9 criteria · 5 companies tracked

Need  4.5
Readiness  4.0
Opportunity  3.3

Business model: Managed AI-as-a-service (SaaS) to enterprise/government

Same need axis as Artificial Intelligence (4.50) but lower opportunity (3.33): India's generative-AI market (\$1.5B) is forecast to grow at roughly half the global rate, even as sovereignty scores a full 5 on the ₹10,371.92 crore IndiaAI Mission.

SCORE BREAKDOWN — ALL 9 CRITERIA

STRATEGIC NEED

- **Sovereignty 5/5:** dedicated national mission with outlay
- **Value-chain 4/5:** input dependency for ≥3 sectors
- **Import substitution:** data gap

READINESS

- **Talent 4/5:** 10k–50k specialists / 8–14 institutions
- **Infrastructure 4/5:** 2–4 operational
- **Cluster deps. 4/5:** cluster needed and exists in India

OPPORTUNITY

- **Global market 4/5:** \$100–500B
- **Domestic market 2/5:** \$500M–2B
- **Growth trajectory 4/5:** 20–30% CAGR

INFRASTRUCTURE & CLUSTER

Sovereignty caveat on the record: the compute is Indian, the silicon is not — H100 hardware is imported [91].

CAPITAL PATHWAY



Common entry point: a government scheme or mission, open to any qualifying company in the sector — not a private deal specific to the named company below.

● **Entry:** IndiaAI Foundation Models pillar — twelve startups selected for sovereign LLM/generative-AI development; the dominant grant-based entry capital in this sector [86]

No sector-specific investor pattern evidenced — Peak XV is a generalist/opportunistic backer here, not a specialist. This is the only named transaction on record for this rung.

● **Bridge:** Sarvam AI \$41M — Lightspeed lead, Peak XV Partners, Khosla Ventures; the largest Series A for an Indian generative-AI startup [87]

No sector-specific investor pattern evidenced — Bessemer is a generalist/opportunistic backer here, not a specialist. This is the only named transaction on record for this rung.

● **Growth:** Sarvam \$234M Series B first close — HCLTech \$150M strategic lead, Bessemer Venture Partners [88]

Not enough evidence to call this a repeatable pattern — this is the only completed exit on record for the sector (India has completed exits in just 30 of 47 tracked sectors).

● **Exit:** Strategic-corporate acquisition — Gupshup's completed acquisitions of AskSid and Active.ai [94]

WHO IS BUILDING

Uniphore — Chennai · Series F · \$870.0M

Yellow.ai — Bengaluru · Series C · \$102.0M

CoRover — Bengaluru · Series A · \$4.0M

SHIPS: **CoRover.ai** — BharatGPT-powered conversational/generative-AI platform that built IRCTC's virtual assistant DISHA, serving 200 · **Yellow.ai** — Dynamic AI Agents (YellowG), a multi-LLM platform used by 1,100+ global enterprises for omnichannel customer · **Verloop.io** — Multilingual (80+ languages) AI customer-support chat and voice automation platform used by 5,000+ businesses

TALENT SOURCING — COMMON PROPERTY

MIXED / NO CLEAR PATTERN

Across seven founders, career stage at founding spans two extremes rather than converging on one pattern: Yellow.ai's Raghu Ravinutala spent 16 years in hardware/semiconductor roles (Texas Instruments, Magma, Broadcom) before founding in his late thirties, while GigaML's Varun Vummadi and Esha Manideep Dinne founded at about 24 straight out of IIT with only internships behind them, and Cogno AI's Aman Goel founded literally as an undergraduate from a class project. Education is equally scattered — NIT Warangal, Jaypee Institute, BITS Pilani, IIT Bombay, IIT Kharagpur, and Verloop.io's Gaurav Singh from a non-elite regional college (Jabalpur Engineering College) — with no institution or career-length recurring often enough to call a pattern; the honest grouping is mid-career hardware/SaaS pivots, serial founders who iterated through prior ventures (Uniphore's Sachdev, Verloop's Singh), and very young IIT graduates founding straight out of research internships. Nasscom estimates a 90% shortage in "GenAI-ready" talent, so the ecosystem-wide figure (120,000+ AI/ML professionals, 185+ CoEs) is the best available proxy for engineering-talent sourcing here.

WHAT TO DO

Investor: Do not underwrite India-domestic GenAI demand at global multiples: the domestic CAGR is 14.59% against a global 29.30% [91][92]. Two rational shapes — (a) export-oriented model/tooling companies priced off the global 29.30% curve, or (b) B2G-anchored...

Government: The Foundation Models pillar has selected twelve teams [90]; the risk now is grant-completion without commercial pull-through. Convert sovereignty spend into demand by routing B2G deployments — the IRCTC DISHA pattern [93] — to grantees.

Corporate: Consolidation is live and cheap relative to the AI sector: three of eight tracked companies have already exited, and Gupshup has closed five deals [94].

Composite 3.62 · 8 of 9 criteria · 10 companies tracked

Need  4.5
Readiness  4.0
Opportunity  2.7

Business model: B2B SaaS subscriptions to industrial enterprises

This is the clearest case of criticality and market size moving independently. Industrial Software scores the maximum 5 on value-chain criticality and only 2 on domestic market size — the widest such spread in this dossier set, and the reason the prioritisation framework is two-axis rather than one.

SCORE BREAKDOWN — ALL 9 CRITERIA

STRATEGIC NEED

- **Sovereignty 4/5:** named in policy with budget line
- **Value-chain 5/5:** global supply ≤3 suppliers/ countries
- **Import substitution:** data gap

READINESS

- **Talent 5/5:** ≥50,000 specialists / ≥15 institutions
- **Infrastructure 3/5:** exactly 1 operational facility
- **Cluster depts. 4/5:** cluster needed and exists in India

OPPORTUNITY

- **Global market 3/5:** \$20–100B
- **Domestic market 2/5:** \$500M–2B
- **Growth trajectory 3/5:** 12–20% CAGR

INFRASTRUCTURE & CLUSTER

India's national answer to the EDA chokepoint is currently access, not ownership: ChipIN centralises access to the triopoly's tools [99][95].

CAPITAL PATHWAY



Common entry point: private angel/VC money, not a government scheme — the named round below is a representative instance of a private-capital-first sector.

● **Entry:** Not a grant-fed deeptech sector — ordinary SaaS venture capital. Hyderabad-based Workroom Automation raised its first institutional round, ₹6.2 crore seed [104] (lead investor name truncated in the evidence record as "Equiru..." — [DATA GAP — needs source re-fetch])

Not enough evidence to establish a common investor pattern for bridge capital here — this is the only named transaction on record.

● **Bridge:** Standard enterprise-SaaS pattern via domestic angel/seed funds — Anscer Robotics raised \$5.4M Series A [105] (lead recorded as "IAN Alpha Fu...", truncated — [DATA GAP — needs source re-fetch])

No completed growth-stage transaction is documented for this sector.

● **Growth:** [DATA GAP — needs a growth-round transaction on record]

Not enough evidence to call this a repeatable pattern — this is the only completed exit on record for the sector (India has completed exits in just 30 of 47 tracked sectors).

● **Exit:** Strategic acquisition by an Indian IT services major expanding digital/AI capability — Wipro Infrastructure Engineering (a Wipro Enterprises subsidiary) [106]

WHO IS BUILDING

Detect Technologies — Chennai · Series B · \$45.7M

SimYog Technology — Bengaluru · Pre-Series A

Entrib Analytics Technology (ShopWorx) — Pune

SHIPS: **Detect Technologies** — T-Pulse, an AI platform (with NOCTUA drone and GUMPS pipeline monitoring) delivering real-time industrial safety · **SimYog Technology** — Compliance-Scope, a virtual-lab EMI/EMC simulation tool letting hardware teams test automotive electronics · **Entrib Analytics (ShopWorx)** — Line MES, a shop-floor Manufacturing Execution System connecting 3,500+ machines across 150+ plant deployments for

TALENT SOURCING — COMMON PROPERTY

THREE FOUNDER PIPELINES (REFINED FROM TWO)

Adding five more companies splits the original two-pattern frame into three. Detect Technologies remains the lone fresh-out-of-college case (four IIT Madras undergraduates). A PhD-researcher pipeline is now confirmed with a second example: SimYog's Dipanjan Gope (IIT Kharagpur, PhD University of Washington, Senior CAD Engineer at Intel, a prior startup exit, then IISc professor before founding) is joined by Eugenie Technologies' Dr. Soudip Roy Chowdhury, who moved from an ML PhD into industry before founding. A third, distinct pipeline of long-tenured corporate/IT-services veterans emerged that wasn't in the original two: ShopWorx's Kiran Nataraj (20+ years across Zensar, Cognizant, Wipro and Persistent Systems) and LivNSense's Avnish Kumar (25+ years in telecom/IoT engineering) both founded after full corporate careers with no academic or fresh-grad profile at all. Only Detect Technologies has documented evidence of an engineering-talent pipeline (IIT Madras student engineers); no other company in the set had public sourcing on where its rank-and-file engineers come from.

WHAT TO DO

Investor: Separate the two theses. (a) Industrial analytics/MES is a normal enterprise-SaaS book — evidenced entry at ₹6.2 crore seed [104], bridge at \$5.4M Series A [105], exit to Indian IT services majors [106] — underwrite it at SaaS multiples against 8.14% domestic...

Government: The mismatch to fix is structural, not budgetary. The DLI sits under a ₹76,000 crore programme [100] and ChipIN gives 100,000 engineers across 400 organisations access to foreign EDA tools [100][95]. Access mitigates the chokepoint; it does not remove it.

Corporate: Wipro's Linecraft.ai acquisition [107] shows Indian IT services majors are the natural acquirer at the analytics layer, and they are buying early.

Composite 3.00 · 9 of 9 criteria · 2 companies tracked



Business model: Hardware product sale to electronics manufacturers

Read those three together: a documented global chokepoint, controlled by one firm, supplying 96% of what India consumes, in a category India has already written a budget line for.

SCORE BREAKDOWN — ALL 9 CRITERIA

STRATEGIC NEED

- **Sovereignty 4/5:** named in policy with budget line
- **Value-chain 5/5:** global supply ≤3 suppliers/ countries
- **Import substitution 4/5:** \$2–10B/yr imported

READINESS

- **Talent 3/5:** 2k–10k / 4–7 institutions
- **Infrastructure 2/5:** funded/announced, not yet operational
- **Cluster deps. 1/5:** needs cluster India does not have

OPPORTUNITY

- **Global market 3/5:** \$20–100B
- **Domestic market 1/5:** <\$500M
- **Growth trajectory 4/5:** 20–30% CAGR

INFRASTRUCTURE & CLUSTER

The missing thing is not money and not people. It is a cluster and an operating fab-to-package pathway . Talent scores 3 — the weakest link is one rung upstream of the founders.

CAPITAL PATHWAY



No entry-capital instrument is documented for this sector — a genuine data gap, not a labeling issue.

● **Entry:** [DATA GAP — needs scheme disbursement records / government grant filings] No dominant entry-capital institution is documented for AI Chips

Fits a broader pattern: Speciale Invest is a thesis-driven deep-tech specialist whose stated mandate explicitly covers this sector — consistent with (though not proof of) a repeatable specialist-investor pathway here.

● **Bridge:** Works — Mindgrove bridged seed to an \$8M Series A co-led by Rocketship.vc and Speciale Invest, with Peak XV Partners participating [211]

Not enough evidence to establish a common investor pattern for growth capital here — this is the only named transaction on record.

● **Growth:** Works, corporate-led — Netrasemi (edge-AI SoC) raised ₹107 crore Series A led by Zoho Corporation with Unicorn India Ventures [212]

Not enough evidence to call this a repeatable pattern — this is the only completed exit on record for the sector (India has completed exits in just 30 of 47 tracked sectors).

● **Exit:** Works, narrowly — L&T Semiconductor Technologies acquired 100% of SiliConch Systems for ~₹183 crore (July 2024) to strengthen its fabless business [213]

WHO IS BUILDING

Only two companies in this sector clear the tracking bar used here — a thin base, and that thinness is part of the argument.

Vervesemi Microelectronics — Greater Noida · Series A

InCore Semiconductors — Chennai · Seed

SHIPS: **Vervesemi Microelectronics** — Precision analog front-end ICs (e.g., 24-bit sigma-delta ADCs for energy metering and medical/industrial sensing) · **InCore Semiconductors** — A six-core RISC-V test chip fabricated on TSMC 40nm using its own Azurite/Calcite core generators, proving a · **Mindgrove Technologies** — Secure IOT (S2401), India's first commercially shipped, indigenously designed RISC-V microcontroller/SoC (700MHz)

TALENT SOURCING — COMMON PROPERTY VETERAN-ANCHORED, SHAKTI-FED

The veteran pattern holds cleanly for Vervesemi Microelectronics, whose two founders — Rakesh Malik (35+ years, Senior Group Manager at STMicroelectronics) and Pratap Narayan Singh (Senior Principal Engineer at STMicroelectronics, 20+ years) — are unambiguous long-tenured semiconductor veterans. InCore Semiconductors complicates a pure veteran read: of its four co-founders, only Chief Architect Gautam Doshi is a classic ~30-year Intel veteran, while CTO Neel Gala, CEO G.S. Madhusudan and Chief Engineer Arjun Menon came straight out of IIT Madras's Shakti open-source RISC-V program with no corporate stint at all. So the genuine pattern is two-layered rather than purely veteran-driven: a senior industry anchor (ST/Intel, 20-30 years) supplies credibility and process rigor, while the founding engineering core is often homegrown out of the domestic Shakti program. That same Shakti program (running at IIT Madras since ~2014, 40+ engineers at its peak spanning students to 50-year veterans) is the confirmed engineering-talent feeder for both InCore's founding team and Mindgrove's core chip IP.

WHAT TO DO

Fund rung one explicitly.

The ₹10,300 crore IndiaAI AI Compute & Semiconductor Infrastructure line already exists [205]. It is a strategy line , not a documented startup entry-capital instrument.

Carve a disclosed design-stage tranche out of it, addressable by fabless startups directly, and publish disbursements — that alone converts a data gap into an observable rung.

Composite 3.11 · 9 of 9 criteria · 27 companies tracked



Business model: Proprietary hardware product sales

The chokepoint here is sharper than in AI chips in one respect: it is a single component. Three firms worldwide make the refrigerator that India's flagship quantum computer runs inside [215], and >90% of critical hardware comes from abroad [214].

SCORE BREAKDOWN — ALL 9 CRITERIA

STRATEGIC NEED

- **Sovereignty 4/5:** named in policy with budget line
- **Value-chain 5/5:** global supply ≤3 suppliers/ countries
- **Import substitution 5/5:** ≥\$10B/yr imported

READINESS

- **Talent 1/5:** <200 / none documented
- **Infrastructure 4/5:** 2–4 operational
- **Cluster deps. 3/5:** cluster exists with documented gaps

OPPORTUNITY

- **Global market 1/5:** <\$5B
- **Domestic market 1/5:** <\$500M
- **Growth trajectory 4/5:** 20–30% CAGR

INFRASTRUCTURE & CLUSTER

152 researchers against a ₹6,003.65 crore mission. That ratio is the sector's binding constraint. NITI Aayog independently names the same shortage in cryogenics, optics and microwave systems [17] — so the deficit is corroborated at both the headcount and the discipline level.

CAPITAL PATHWAY



Common entry point: a government scheme or mission, open to any qualifying company in the sector — not a private deal specific to the named company below.

● **Entry:** Works, but intermediated — under NQM guidelines, T-Hubs provide initial seed backing (Track-1, up to ₹5 crore, in exchange for equity) before private VC/PE follows [224]

Not enough evidence to establish a common investor pattern for bridge capital here — this is the only named transaction on record.

● **Bridge:** Works, and the mission co-invests directly — QpiAI's \$32M Series A was led jointly by Avataar Ventures and the National Quantum Mission (DST) [[225], [226]]

Not enough evidence to establish a common investor pattern for growth capital here — this is the only named transaction on record.

● **Growth:** Works — the Technology Development Board (DST) offers grants, loans and equity to quantum startups beyond the T-Hub seed/ Series-A stage [227]

No common exit pattern: this is one of the majority of sectors with zero completed exits on record. Ecosystem-wide, 0 of 516 investors profiled report any exit data — the single biggest caveat on every route in this report.

● **Exit:** [DATA GAP — needs M&A filings / exchange disclosures] No realistic exit route documented

WHO IS BUILDING

27 companies tracked — by far the largest base of the four compute sectors, and a genuine surprise against a readiness score of 2.67.

QpiAI — Bengaluru · Series A

QNu Labs — Bengaluru · Series A

BosonQ Psi (BQP) — Bengaluru · Seed

SHIPS: **QpiAI** — QpiAI Indus, India's first full-stack 25-qubit superconducting quantum computer, plus the 64-qubit QpiAI Kaveri · **QNu Labs** — Armos (quantum key distributor) and Tropos (quantum random number generator), commercial products listed on · **BosonQ Psi** — BQPhy, described as the world's first quantum-powered engineering simulation SaaS product, for automotive and

TALENT SOURCING — COMMON PROPERTY **VETERAN-LED AT SCALE, ACADEMIC-SPINOUT LONGTAIL**

The corporate/PSU veteran pattern holds cleanly only at the two largest, most-funded companies: QpiAI (founder Dr. Nagendra Nagaraja, confirmed ex-NVIDIA, ex-Qualcomm) and QNu Labs (CEO Sunil Gupta, ex-C-DOT, Infosys, Paladion) — both build full-stack, enterprise/ government-sold platforms. Below that tier the picture flips to academic spinouts: Quanastra's founder Dr. Vidur Raj went directly from an Australian National University postdoc to founding the company, QuPrayog was co-founded by a sitting IIT Madras professor, Dimira Technologies pairs an IIT Bombay professor with an industry-operations director, and PrenishQ describes itself as run by research students and scientists out of its IIT Delhi incubation. So the veteran-CEO profile is specific to the handful of firms selling integrated computing or security platforms, while the hardware/photonics layer of the sector runs closer to academic norm. Engineering talent still tracks the National Quantum Mission's academic Thematic Hub structure (IISc=computing, IIT Madras=communication, IIT Bombay=sensing/metrology, IIT Delhi=materials), though which institution incubated a startup matters more in practice than its assigned technology theme.

WHAT TO DO

Treat talent as the mission's binding constraint, not a side programme. 152 researchers [216] against a ₹6,003.65 crore outlay [216] is a spending-capacity problem before it is a science problem.

Fund the specific disciplines NITI Aayog names — cryogenics, optics, microwave systems, techno-business [17]. Widen the T-Hub gate.

If Track-1 equity at up to ₹5 crore is the only entry door [224], hub capacity caps sector formation.

Composite 3.75 · 8 of 9 criteria · 3 companies tracked



Business model: Full-stack GPU/compute cloud services

The sovereignty argument is the strongest of the four sectors here: 44% of a national AI mission's money is allocated to this one pillar [228].

SCORE BREAKDOWN — ALL 9 CRITERIA

STRATEGIC NEED

- **Sovereignty 5/5:** dedicated national mission with outlay
- **Value-chain 4/5:** input dependency for ≥3 sectors
- **Import substitution:** data gap

READINESS

- **Talent 4/5:** 10k–50k specialists / 8–14 institutions
- **Infrastructure 4/5:** 2–4 operational
- **Cluster deps. 3/5:** cluster exists with documented gaps

OPPORTUNITY

- **Global market 4/5:** \$100–500B
- **Domestic market 3/5:** \$2–10B
- **Growth trajectory 3/5:** 12–20% CAGR

INFRASTRUCTURE & CLUSTER

The precise deficit is power-and-facility fitness for AI workloads, not compute access. 1,123 MW of data centre capacity exists [229]; the documented gap is that this capacity was not built for AI-class density.

CAPITAL PATHWAY



Common entry point: a government scheme or mission, open to any qualifying company in the sector — not a private deal specific to the named company below.

● **Entry:** Works, as infrastructure-in-kind — the IndiaAI Compute Pillar's subsidised GPU access functions as government infrastructure-as-entry-capital: ₹65/hour, 38,000 GPUs [86]

No sector-specific investor pattern evidenced — Nexus Venture Partners is a generalist/opportunistic backer here, not a specialist. This is the only named transaction on record for this rung.

● **Bridge:** Works — Neysa's Series A co-led by NTTVC, Z47 (formerly Matrix Partners India) and Nexus Venture Partners [233]

No sector-specific investor pattern evidenced — Nexus Venture Partners is a generalist/opportunistic backer here, not a specialist. This is the only named transaction on record for this rung.

● **Growth:** Works, at global scale — Neysa's \$1.2 billion raise led by Blackstone, with Teachers' Venture Growth, TVS Capital, 360 ONE Assets and Nexus Venture Partners as equity co-investors [234]

Not enough evidence to call this a repeatable pattern — this is the only completed exit on record for the sector (India has completed exits in just 30 of 47 tracked sectors).

● **Exit:** Works — strategic acquisition by domestic IT/fintech-software firms: Aurionpro acquired 67% of Arya.ai (Lithasa Technologies) for ~\$16.5M / ₹135.20 crore, April 2024 [235]

WHO IS BUILDING

Four companies tracked — a small base for a \$3.8B market [204], and a signal that this sector's capital is concentrating in a few very large balance sheets rather than dispersing across many startups.

Yotta Data Services (Shakti Cloud) — Neysa — EdgeNeural AI —

SHIPS: **Yotta Data Services** — Shakti Cloud — India's largest AI supercomputer, built with NVIDIA H100 GPUs (scaling toward 32,768 GPUs) · **Neysa** — Velocis AI Cloud — GPU-as-a-Service on NVIDIA/AMD bare metal, VM or Kubernetes, plus an AI-Platform-as-a-Service · **Portkey** — An AI Gateway and LLMops observability platform routing to 1,600+ LLMs across 60+ providers, processing 500B+

TALENT SOURCING — COMMON PROPERTY TWO TRACKS: INFRA VETERANS VS AI-TOOLING FOUNDERS

The infrastructure-operator pattern holds firmly for the neo-cloud/GPU-infrastructure companies: Yotta's Sunil Gupta (NIT Kurukshetra engineering degree, Columbia MBA) spent 30+ years building and running data centres, including nine years as President of NTT Netmagic (2010-2019), before launching Shakti Cloud, while Neysa's Sharad Sanghi and Anindya Das ran that same Netmagic-to-NTT operation together (Sanghi as CEO, Das as CTO) before spinning out their own GPU cloud in 2023. But companies also tagged 'AI Infrastructure' by deal trackers split into a second, unrelated founder population: Portkey's Rohit Agarwal spent roughly a decade building AI products (Framebench, acquired by Freshworks; then Pepper) before founding an LLM-gateway/observability startup, and EdgeNeural.ai was founded in 2020 by young engineers with no long prior corporate career, building edge-AI deployment tooling. So 'AI Infrastructure' is really two distinct founder tracks — decades-long data-centre operators building physical GPU cloud, and shorter-tenured AI-software builders building the tooling layer on top of it. Engineering-talent sourcing is still best evidenced by the ecosystem-wide GCC figure (120,000+ AI/ML professionals, 185+ COEs), though Neysa and Yotta both visibly draw operations engineers from the same Netmagic/NTT alumni bench their founders came from.

WHAT TO DO

Spend the pillar, and show the spend. ₹4,563.36 crore is allocated to the Compute pillar [228]; the observable instrument to date is the ₹65/hour GPU subsidy at 38,000 GPUs [86].

Attack the AI-workload gap in the data centre cluster, not the GPU count.

GPU access is solved at ₹65/hour [86]; the 1,123 MW base with documented AI-workload gaps [229] is not.

Composite 3.67 · 9 of 9 criteria · 6 companies tracked

Need  **4.7**
Readiness  **3.0**
Opportunity  **3.3**

Business model: Contract/EMS-ODM manufacturing for third-party brands

\$36.8B of annual component imports [236] is, by a wide margin, the largest single import-substitution figure among these four sectors — more than ten times the \$3.4B AI-semiconductor figure [204]. Against it, a ~\$2.7B scheme [236].

SCORE BREAKDOWN — ALL 9 CRITERIA

STRATEGIC NEED

- **Sovereignty 5/5:** dedicated national mission with outlay
- **Value-chain 4/5:** input dependency for ≥3 sectors
- **Import substitution 5/5:** ≥\$10B/yr imported

READINESS

- **Talent 2/5:** 200–2,000 / 1–3 institutions
- **Infrastructure 4/5:** 2–4 operational
- **Cluster deps. 3/5:** cluster exists with documented gaps

OPPORTUNITY

- **Global market 3/5:** \$20–100B
- **Domestic market 4/5:** \$10–50B
- **Growth trajectory 3/5:** 12–20% CAGR

INFRASTRUCTURE & CLUSTER

The deficit is specific and unusual: the factories are operational and the research institutions are not there. Five EMCs are running [240]; the R&D base sits at three institutions [238].

CAPITAL PATHWAY



Common entry point: a government scheme or mission, open to any qualifying company in the sector — not a private deal specific to the named company below.

● **Entry:** Works, scheme-led — ECMS, a ₹22,919 crore PLI scheme for electronic component manufacturing, alongside SPECS [247]

No sector-specific investor pattern evidenced — Nexus Venture Partners is a generalist/opportunistic backer here, not a specialist. This is the only named transaction on record for this rung.

● **Bridge:** Works, thinly — Elecbits (electronics design and manufacturing) raised a Series A led by Nexus Venture Partners [248]

Not enough evidence to establish a common investor pattern for growth capital here — this is the only named transaction on record.

● **Growth:** Works, strategic — Applied Ventures, the VC arm of Applied Materials, made a growth equity investment in VVDN Technologies [249]

Not enough evidence to call this a repeatable pattern — this is the only completed exit on record for the sector (India has completed exits in just 30 of 47 tracked sectors).

● **Exit:** Works — Tata Companies acquired a 100% indirect stake in Wistron's India iPhone manufacturing unit for \$125 million [250]

WHO IS BUILDING

Six tracked companies — and unlike the other three sectors, three of them are already public. This is the one compute sector where the domestic industrial base, not the startup base, is the story.

Dixon Technologies (India) —

Data Patterns (India) —

Syrma SGS Technology —

SHIPS: **Dixon Technologies** — EMS/ODM contract manufacturing — produces Google Pixel smartphones (for export to US/Europe), Nokia-brand phones · **Data Patterns (India)** — BrahMos Missile Checkout Equipment built for DRDO/BrahMos Aerospace, plus avionics/electronic-warfare systems for · **Syrma SGS Technology** — EMS/PCB-assembly and RFID-tag manufacturing serving automotive, consumer-electronics, and medical-device OEMs

TALENT SOURCING — COMMON PROPERTY VETERAN OPERATORS, COMPRESSED FOR NEWER ENTRANTS

The veteran-operator pattern strengthens with more evidence: Dixon Technologies' Sunil Vachani (32 years in electronics manufacturing, a second-generation entrant into his father's TV business) and Syrma SGS's Sandeep Tandon (USC-trained electrical engineer, 18+ years running EMS supplier Celetronix before founding Syrma in 2004) both spent one-and-a-half to three decades as manufacturing operators before/while building their companies, matching Data Patterns' Rangarajan (IIT Madras M.Sc, growing his own defence-electronics company since 1985). The newest entrant, TIEA Connectors (founded 2020), compresses the timeline but not the underlying principle: CEO Ajith Sasidharan spent about a decade at a connector manufacturer (Ucore Connectronics) and infrastructure firms before founding — shorter tenure, but still directly domain-relevant. The common thread is direct, relevant manufacturing-floor experience before founding, with required tenure shrinking as the sector matures. Engineering-talent sourcing is still best evidenced by the Karnataka MNC-R&D-centre figure (400+ MNC R&D centres, 85+ chip-design houses), supplemented by TIEA's own founder path — a lateral move from one connector/EMS manufacturer straight into another — suggesting engineers in this sector also move company-to-company within the same manufacturing niche rather than entering fresh from campus.

WHAT TO DO

Point ECMS at bare PCBs specifically.

The scheme is ₹22,919 crore / ~\$2.7B [[247], [236]]; the named cluster gap is that Sriperumbudur and Noida still import the large majority of bare-PCB demand [239].

Build the R&D institution base, because that is the criterion at 2.

Composite 3.75 · 8 of 9 criteria · 30 companies tracked

Need  **5.0**
Readiness  **3.3**
Opportunity  **3.3**

Business model: Vertically integrated B2C hardware + charging

The strongest argument here is the chokepoint, not the market. CRISIL's framing is the one a policymaker should internalise: rare-earth magnets are less than 5% of a vehicle's cost yet indispensable to EV motors and electric steering [261].

SCORE BREAKDOWN — ALL 9 CRITERIA

STRATEGIC NEED

- **Sovereignty 5/5:** dedicated national mission with outlay
- **Value-chain 5/5:** global supply ≤3 suppliers/countries
- **Import substitution:** data gap

READINESS

- **Talent 3/5:** 2k–10k / 4–7 institutions
- **Infrastructure 4/5:** 2–4 operational
- **Cluster depts. 3/5:** cluster exists with documented gaps

OPPORTUNITY

- **Global market 3/5:** \$20–100B
- **Domestic market 3/5:** \$2–10B
- **Growth trajectory 4/5:** 20–30% CAGR

INFRASTRUCTURE & CLUSTER

The named missing thing is not the assembly line. India can assemble. The missing thing is the magnet-to-motor step and the certification throughput to qualify domestic powertrain parts at scale.

CAPITAL PATHWAY



Common entry point: private angel/VC money, not a government scheme — the named round below is a representative instance of a private-capital-first sector.

● **Entry:** Founder-network angel — Ather's first seed from Flipkart founders [268]

No sector-specific investor pattern evidenced — Tiger Global is a generalist/opportunistic backer here, not a specialist. This is the only named transaction on record for this rung.

● **Bridge:** Global growth VC — Tiger Global's \$12M into Ather [268]

Not enough evidence to establish a common investor pattern for growth capital here — this is the only named transaction on record.

● **Growth:** PLI-Auto mission money, not late-stage venture equity [269]

No common exit pattern: this is one of the majority of sectors with zero completed exits on record. Ecosystem-wide, 0 of 516 investors profiled report any exit data — the single biggest caveat on every route in this report.

● **Exit:** [DATA GAP]

WHO IS BUILDING

Ola Electric — Bengaluru · IPO · \$1000.0M

Ather Energy — Bengaluru · IPO · \$502.0M

Ultraviolette — Bengaluru · Series E · \$100.0M

SHIPS: **Ola Electric** — The S1 electric-scooter line (S1 Air/X/Pro) and the Roadster electric-motorcycle series, mass-manufactured at the · **Ather Energy** — The Ather 450 scooter series and Ather Rizta family scooter, sold with the proprietary Ather Grid fast-charging · **Vecmocon Technologies** — A CAN-enabled Battery Management System (BMS) deployed across 120,000+ units and 100+ OEM customers (e.g., BGauss)

TALENT SOURCING — COMMON PROPERTY YOUNG FOUNDERS, MINIMAL PRIOR CAREER

The 'young IIT founders straight into founding' pattern holds and gains a third variant: alongside Ola Electric's Bhavish Aggarwal (IIT Bombay 2008, a brief stint at Microsoft Research India before serial-founding) and Ather's Tarun Mehta (IIT Madras, quit an Ashok Leyland job to found Ather in 2013), Vecmocon Technologies shows a campus-incubation route into the same young-founder pattern — the company was incubated directly inside IIT Delhi in 2016–17 on DST and institute grants, taking its Battery-Management-System research straight from the lab into a company with no intervening corporate career at all. Ultraviolette Automotive fits the same age/sector profile, though its founders' pre-founding careers are not independently documented. Across the well-documented companies, founders are uniformly young, IIT-pedigreed, and short on (or entirely without) prior corporate tenure — the only variation is the on-ramp: serial-founding, quitting an auto-industry job, or a direct campus tech-transfer. Engineering talent draws from the same two pools visible in the founders: auto-industry incumbents (Ashok Leyland) and campus-incubation programs like IIT Delhi's, which fed Vecmocon's own founding team.

WHAT TO DO

Attack the magnet step, not the vehicle.

The chokepoint is documented at >90% Chinese processing concentration [261] on an input worth <5% of vehicle cost [261].

Convert PM E-DRIVE from demand-side to qualification-side.

Composite 3.67 · 9 of 9 criteria · 22 companies tracked



Business model: B2B cell/pack supply to commercial EV OEMs

BatteryTech is the strongest-evidenced case in the family: an upstream chokepoint, a funded mission, and a quantified import bill. Roughly \$3B a year leaves India for cells that a domestic industry is licensed to build [273].

SCORE BREAKDOWN — ALL 9 CRITERIA

STRATEGIC NEED

- **Sovereignty 5/5:** dedicated national mission with outlay
- **Value-chain 5/5:** global supply ≤3 suppliers/countries
- **Import substitution 4/5:** \$2–10B/yr imported

READINESS

- **Talent 3/5:** 2k–10k / 4–7 institutions
- **Infrastructure 3/5:** exactly 1 operational facility
- **Cluster deps. 3/5:** cluster exists with documented gaps

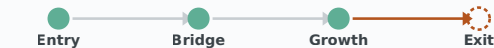
OPPORTUNITY

- **Global market 4/5:** \$100–500B
- **Domestic market 3/5:** \$2–10B
- **Growth trajectory 3/5:** 12–20% CAGR

INFRASTRUCTURE & CLUSTER

The named missing thing: the 39 GWh gap between award and installation. Every other deficit in this sector is downstream of it. Note also that the 72% cluster concentration figure is a share of announced capacity — a map of intentions, not of output.

CAPITAL PATHWAY



Common entry point: private angel/VC money, not a government scheme — the named round below is a representative instance of a private-capital-first sector.

● **Entry:** Angel/family-office seed — Xbattery's \$2.3M seed [281]

Not enough evidence to establish a common investor pattern for bridge capital here — this is the only named transaction on record.

● **Bridge:** Corporate-strategic anchor — Amara Raja Batteries anchoring Log9 Materials' Series A+ [282]

Not enough evidence to establish a common investor pattern for growth capital here — this is the only named transaction on record.

● **Growth:** PLI ACC mission money, not late-stage venture equity [283]

No common exit pattern: this is one of the majority of sectors with zero completed exits on record. Ecosystem-wide, 0 of 516 investors profiled report any exit data — the single biggest caveat on every route in this report.

● **Exit:** [DATA GAP]

WHO IS BUILDING

- LOHUM** — Noida · Series B · \$90.0M **Exponent Energy** — Bengaluru · Series B · \$65.7M
- Cygni Energy** — Hyderabad · Series A · \$6.4M

SHIPS: **LOHUM** — Battery recycling and critical-materials recovery (lithium, cobalt, nickel) from spent EV/industrial batteries · **Exponent Energy** — The e^pump 15-minute rapid-charging energy stack (battery + charger), deployed via OEM partners like Montra and · **Arkle Energy Solutions** — Vanadium redox-flow batteries for long-duration, grid-scale energy storage (company confirmed live; specific

TALENT SOURCING — COMMON PROPERTY SAME PEDIGREE, DIFFERENT TENURE

The 'same IIT pedigree, different clock' finding holds: LOHUM's path from IIT Kanpur/Harvard MBA straight to founding (now running a critical-materials/battery-recycling business) contrasts with Exponent Energy's Arun Vinayak, who spent seven years inside Ather Energy's electric-scooter business before founding his own rapid-charging company. Beyond these two, public founder-background detail is scarce across the rest of the sector's named companies (Cygni Energy, Arkle Energy Solutions) — real, funded businesses, but without a documented founder story to compare against LOHUM and Exponent. So the pattern is real but the evidence base behind it stays thin: two companies, two different career lengths, one shared campus pedigree. Engineering-talent sourcing continues to point to prior EV-industry incumbents, per the finding that Exponent's founder built his battery expertise inside Ather's vertically integrated two-wheeler ecosystem before founding his own company.

WHAT TO DO

- Publish and enforce the installation calendar. 1 GWh of 40 GWh awarded is the single most decision-relevant number in this dossier [253].
- Sequence the mineral mission ahead of the cell mission.
- The chokepoint is refining, not assembly [271]. The National Critical Mineral Mission's ₹16,300 cr over 7 years covering 30 minerals including lithium and cobalt [254] must land refining capacity before the 50 GWh of...

Composite 3.11 · 9 of 9 criteria · 15 companies tracked

Need  **4.0**
Readiness  **2.7**
Opportunity  **2.7**

Business model: PE-backed domestic electrolyser manufacturing

The demand case is already physical, not speculative: India's fertiliser industry alone consumes 5–6 MMT of grey (fossil-based) hydrogen annually [259]. Green hydrogen's addressable market is a substitution market that already exists at scale.

SCORE BREAKDOWN — ALL 9 CRITERIA

STRATEGIC NEED

- **Sovereignty 5/5:** dedicated national mission with outlay
- **Value-chain 4/5:** input dependency for ≥3 sectors
- **Import substitution 3/5:** \$500M–2B/yr imported

READINESS

- **Talent 3/5:** 2k–10k / 4–7 institutions
- **Infrastructure 2/5:** funded/announced, not yet operational
- **Cluster deps. 3/5:** cluster exists with documented gaps

OPPORTUNITY

- **Global market 2/5:** \$5–20B
- **Domestic market 1/5:** <\$500M
- **Growth trajectory 5/5:** ≥30% CAGR

INFRASTRUCTURE & CLUSTER

This is the whole readiness story: India has awarded 3,000 MW of electrolyser manufacturing and operates 10 Nm³/hr pilots. "Announced" and "operational" are different claims. The rubric scores the second, which is why Hydrogen reads 2 on infrastructure while its outlay reads 5 on sovereignty.

CAPITAL PATHWAY



Common entry point: a government scheme or mission, open to any qualifying company in the sector — not a private deal specific to the named company below.

- **Entry:** Non-dilutive government R&D funding via DST's Hydrogen & Fuel Cell programme, not startup equity [289]

No sector-specific investor pattern evidenced — Peak XV is a generalist/opportunistic backer here, not a specialist. This is the only named transaction on record for this rung.

- **Bridge:** Newtrace's Pre-Series A, led by HDFC Bank and Mitsui Sumitomo Insurance Venture Capital, with Peak XV's Surge and Aavishkaar Capital participating [290]

Not enough evidence to establish a common investor pattern for growth capital here — this is the only named transaction on record.

- **Growth:** SIGHT-tranche PLI awarded to large corporates (RIL, Adani, L&T) — not venture equity to startups [291]

No common exit pattern: the sector's own evidence explicitly documents that no completed acquisition or listing exists yet, consistent with the ecosystem-wide finding that 0 of 516 investors profiled report exit data.

- **Exit:** No completed exit in India's green hydrogen deep-tech sector; the fund's broader finding is India has yet to see a large exit for any Indian deep-tech company [292]

WHO IS BUILDING

Newtrace — Bengaluru · Pre-Series A · \$6.3M

H2Loop — Bengaluru · Seed · \$2.0M

h2e Power — Pune

SHIPS: **h2e Power** — Electrolysers across four technologies, including a 500 kW anion-exchange-membrane (AEM) electrolyser, sold into · **Newtrace** — Developing a lower-cost electrolyser design for green-hydrogen production; exact specifications are not yet public

TALENT SOURCING — COMMON PROPERTY THIN EVIDENCE

This sector remains genuinely thin: Newtrace's domain has lapsed and H2Loop returned no verifiable founder information, leaving the original finding intact — Newtrace's two documented founders are PhD-trained researchers (aerospace and applied physics backgrounds) who moved directly from research into founding, with no long corporate career in between. h2e Power sits at the opposite extreme: founder Siddharth Mayur is a self-described 'first-generation social entrepreneur' with a business/clean-energy-ventures background rather than a technical degree, and its engineering leadership is a veteran industrial electrical engineer rather than a fresh graduate or national-lab alumnus. The two data points do not share a pattern — one is academic-research-to-founder, the other non-technical-entrepreneur-plus-hired-engineering-veteran — so this remains a genuinely thin, non-generalizable sector rather than one with a hidden common thread, and no sector-specific engineering-talent evidence beyond the h2e Power example could be found.

WHAT TO DO

Report SIGHT against commissioning, not award. 3,000 MW across 15 firms is an allocation [259]; the operational estate is port pilots at Kandla and V.O. Chidambaranar [256].

Fund the membrane and the catalyst.

The documented cluster gaps are Nafion membranes and PGM catalysts [285]. A ₹19,744 cr mission [256] that scales electrolyser assembly on imported membranes reproduces the battery sector's cathode problem one layer up.

Composite 3.75 · 8 of 9 criteria · 66 companies tracked



Business model: Direct B2B product sale, compliance-driven demand

Sovereignty fired on the existence of a dedicated mission architecture (NAPCC), not a stated budget. ClimateTech's own scoring overlaps EnergyTech, BatteryTech, Hydrogen and EV Technology — each scored separately — so don't double-count across them.

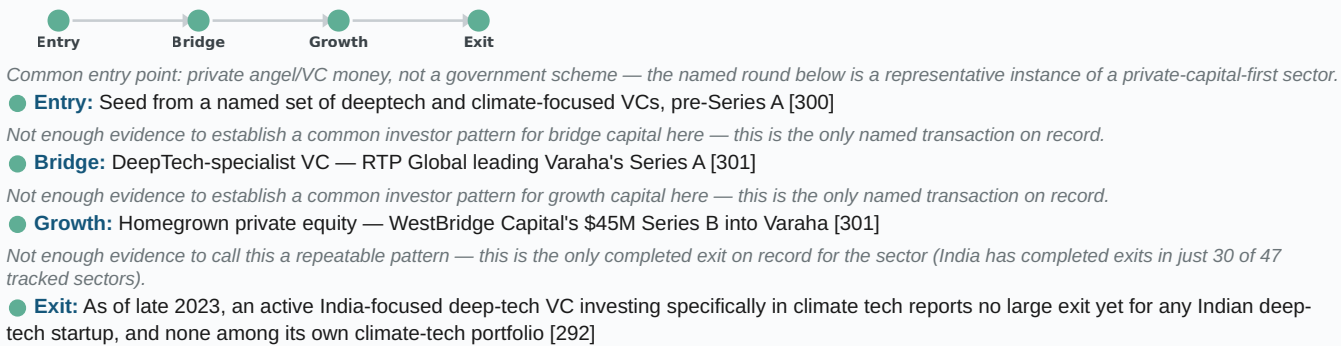
SCORE BREAKDOWN — ALL 9 CRITERIA

STRATEGIC NEED	READINESS	OPPORTUNITY
● Sovereignty 5/5: dedicated national mission with outlay ● Value-chain 4/5: input dependency for ≥3 sectors 🔗 Import substitution: data gap	● Talent 3/5: 2k–10k / 4–7 institutions ● Infrastructure 4/5: 2–4 operational ● Cluster deps. 4/5: cluster needed and exists in India	● Global market 3/5: \$20–100B ● Domestic market 3/5: \$2–10B ● Growth trajectory 4/5: 20–30% CAGR

INFRASTRUCTURE & CLUSTER

The named missing thing is not infrastructure or clusters — those are the family's best. It is talent depth (3) and an unmeasured import position . ClimateTech is the one sector here where the deficit is human and informational rather than physical.

CAPITAL PATHWAY



WHO IS BUILDING

Varaha — Gurugram · Series B · \$12.7M	IBISA Network — Luxembourg · Seed · \$4.7M	Chakr Innovation — Gurugram · Series C
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SHIPS: **Varaha** — MRV platform combining remote sensing and ML to certify carbon credits from regenerative agriculture, biochar and · **IBISA Network** — Satellite/index-based parametric heat-stress insurance for dairy farmers in India and Bangladesh, paying out · **Chakr Innovation** — Chakr Shield, a diesel-generator retrofit that captures particulate emissions and converts soot into usable carbon

TALENT SOURCING — COMMON PROPERTY NO SINGLE PATH — MIXED BACKGROUNDS

Across five ClimateTech companies, founder backgrounds show no single common path: Varaha's Madhur Jain built AgriTech/social-enterprise operating experience before founding; Chakr Innovation's Kushagra Srivastava and Arpit Dhupar went straight from the IIT Delhi campus into founding with no prior job; GPS Renewables' Mainak Chakraborty spent years as a corporate engineer (Tata Elxsi) and then an entrepreneur-in-residence before founding; and Uravu Labs' founders moved directly from NIT Calicut student water-scarcity projects into the company. IBISA Network's Maria Mateo Iborra is the outlier — a two-decade satellite-communications industry veteran, though the company is Luxembourg-headquartered with India only as a market. Engineering talent still traces most visibly to IIT/NIT programs: Chakr's IIT Delhi-origin team has scaled to roughly 200 engineers/scientists, and Uravu's founding core shares an NIT Calicut alumni base.

WHAT TO DO

Put an outlay on the mission.

NAPCC's 8 National Missions are anchored under ministries but carry no stated outlay [255] — the only sovereignty-5 in this family without a rupee figure.

A published, ministry-wise NAPCC budget line is prerequisite to any private co-investment case.

Composite 3.33 · 9 of 9 criteria · 38 companies tracked

Need  **4.0**
Readiness  **3.3**
Opportunity  **2.7**

Business model: Patented-material manufacturing via OEM partnerships

The two criteria scoring 5 tell the story: Semiconductors, BatteryTech and EV are all downstream of a magnet/rare-earth chokepoint India does not control. The \$221M import figure is small because India buys magnets, not because the dependency is.

SCORE BREAKDOWN — ALL 9 CRITERIA

STRATEGIC NEED

- **Sovereignty 5/5:** dedicated national mission with outlay
- **Value-chain 5/5:** global supply ≤3 suppliers/countries
- **Import substitution 2/5:** \$50–500M/yr imported

READINESS

- **Talent 3/5:** 2k–10k / 4–7 institutions
- **Infrastructure 4/5:** 2–4 operational
- **Cluster deps. 3/5:** cluster exists with documented gaps

OPPORTUNITY

- **Global market 3/5:** \$20–100B
- **Domestic market 3/5:** \$2–10B
- **Growth trajectory 2/5:** 6–12% CAGR

INFRASTRUCTURE & CLUSTER

The missing thing has a name: heavy rare-earth separation and refining capacity. Not talent, not synchrotron time, not mission money. One unbuilt process step.

CAPITAL PATHWAY



Common entry point: private angel/VC money, not a government scheme — the named round below is a representative instance of a private-capital-first sector.

● **Entry:** Working — specialist early-stage VC, not grants; Theia Ventures led a pre-seed in AI material discovery [313]

No sector-specific investor pattern evidenced — Lightspeed is a generalist/opportunistic backer here, not a specialist. This is the only named transaction on record for this rung.

● **Bridge:** Working — Whizzo raised a \$15M Series A led by Fundamentum Partnership with LB Investment and Lightspeed [314]

Not enough evidence to establish a common investor pattern for growth capital here — this is the only named transaction on record.

● **Growth:** Being built by the state — NCMM is the growth-stage co-investment channel for critical-minerals processing [315]

No common exit pattern: the sector's own evidence explicitly documents that no completed acquisition or listing exists yet, consistent with the ecosystem-wide finding that 0 of 516 investors profiled report exit data.

● **Exit:** BROKEN — no completed acquisition of an Indian advanced-materials startup identified; Tata Steel and Aditya Birla are documented as potential but unproven acquirers [314]

WHO IS BUILDING

38 companies tracked. Log9 Materials (Bengaluru, Series A, \$60M) ships the RapidX battery series for two- and three-wheeler EVs [311] and monetises through OEM integration — its Musashi Seimitsu partnership is the documented model: patented material, sold through a strategic manufacturer [311].

SHIPS: **Log9 Materials** — RapidX battery series for two- and three-wheeler EVs, offering fast charging and extended lifespans, integrated · **Newtrace** — Voltagen, a membrane-less electrode/stack technology for alkaline water electrolyzers cutting component count ~70% · **Carbon Craft Design** — Carbon Tile, a building tile made by capturing airborne carbon/pollution particulates and fusing them with cement

TALENT SOURCING — COMMON PROPERTY RESEARCH-BACKGROUND LEANS, NOT UNIFORM

Adding three companies (Newtrace, Carbon Craft Design, ThinkMetal) to Log9 Materials and Whizzo shows a lean rather than a strict rule: three of five founders — Log9's Akshay Singhal (IIT Roorkee student researcher), Newtrace's Prasanta Sarkar and Rochan Sinha (PhD/fellowship-level electrochemistry researchers), and Carbon Craft Design's Tejas Sidnal (independent design-research background) — moved from academic or research work fairly directly into founding without a long corporate career. That pattern breaks for Whizzo (founder was a Zetwerk corporate executive) and ThinkMetal (Sabyasachi Ghosh and Arushi Sharma, SRM-trained mechanical engineers who met on a Hyperloop competition team and did R&D at another 3D-printing startup before founding). Engineering talent for harder materials science leans on national research-institute facilities beyond any one founder's home campus — Log9 runs a formal collaboration with IISc's Centre for Nano Science and Engineering, and Newtrace's 45-plus-engineer Bengaluru technology center draws visibly from IIT Bhubaneswar and IIT Madras alumni.

WHAT TO DO

Deploy the ₹16,300 crore direct-expenditure tranche of NCMM against the named gap, not the whole chain — heavy rare-earth separation and refining, which is the one segment documented as absent [302][303].

Three separation plants already run far below rated capacity [303]. Convert the ₹18,000 crore expected PSU component into an offtake instrument [302] — PSU purchase commitments substitute for the missing exit by giving...

Composite 3.67 · 9 of 9 criteria · 9 companies tracked



Business model: Proprietary laser/optical-fiber hardware products

The missing thing is a supplier base, not a knowledge base. Eight-plus research institutions and four national facilities sit above a precision-optics manufacturing layer that reduces, in the evidence, to a handful of named firms [322][317][321].

SCORE BREAKDOWN — ALL 9 CRITERIA

STRATEGIC NEED

- **Sovereignty 4/5:** named in policy with budget line
- **Value-chain 4/5:** input dependency for ≥3 sectors
- **Import substitution 5/5:** ≥\$10B/yr imported

READINESS

- **Talent 4/5:** 10k–50k specialists / 8–14 institutions
- **Infrastructure 4/5:** 2–4 operational
- **Cluster deps. 3/5:** cluster exists with documented gaps

OPPORTUNITY

- **Global market 4/5:** \$100–500B
- **Domestic market 2/5:** \$500M–2B
- **Growth trajectory 3/5:** 12–20% CAGR

INFRASTRUCTURE & CLUSTER

The missing thing is a supplier base, not a knowledge base. Eight-plus research institutions and four national facilities sit above a precision-optics manufacturing layer that reduces, in the evidence, to a handful of named firms.

CAPITAL PATHWAY



Common entry point: a government scheme or mission, open to any qualifying company in the sector — not a private deal specific to the named company below.

● **Entry:** Working (public) — DRDO's Technology Development Fund brings MSMEs and startups into defence and dual-use R&D including photonics/laser systems [330][331]

Fits a broader pattern: pi Ventures is a thesis-driven deep-tech specialist whose stated mandate explicitly covers this sector — consistent with (though not proof of) a repeatable specialist-investor pathway here.

● **Bridge:** Working (thin) — LightSpeed Photonics raised \$6.5M pre-Series A led by pi Ventures, with 500 Global, Indian Accelerator, 8X Ventures, Java Capital [331][332]

Not enough evidence to establish a common investor pattern for growth capital here — this is the only named transaction on record.

● **Growth:** BROKEN — no pool exists. Of 50 tracked photonics startups, only 17 are funded and only 3 have secured Series A or later [333]

No common exit pattern: the sector's own evidence explicitly documents that no completed acquisition or listing exists yet, consistent with the ecosystem-wide finding that 0 of 516 investors profiled report exit data.

● **Exit:** Absent — no completed acquisition of an Indian photonics startup found across Indian business press, Tracxn and photonics M&A trackers [334]; listed players (Bharat Electronics, Dixon, Sterlite, Birla Cable) are diversified conglomerates rather than pure-play consolidators [334]

WHO IS BUILDING

Nine tracked companies, three with recorded funding: LightSpeed Photonics (Hyderabad, \$8.5M), olee.space (Pune, seed, \$3M), Quanfluence (Bengaluru, \$2M). Also Velmenni, Teralumen Solutions, Tractrix Opto Dynamics, Gyroid Photonics, Aksh Optifibre, Lumisoft Technologies.

SHIPS: DeepLase — Narrow-linewidth fiber lasers, single-photon sources and specialty optical fibers for quantum tech, telecom · olee.space — Free-space optical communication (FSOC) laser links — demonstrated a 20km/10Gbps indigenous wireless laser link — · **Quanfluence** — OPTIQON, a hardware-accelerated coherent Ising machine/photonic optimization platform, plus standalone quantum

TALENT SOURCING — COMMON PROPERTY

VETERAN-PIVOT + ACADEMIC SPINOUT

Three additional real, India-connected photonics companies sit alongside the original single data point, sharpening rather than overturning the earlier finding: the strongest pattern is veteran semiconductor/photonics professionals with a decade or more of industry experience pivoting into deep-tech photonics as second-time founders — Quanfluence's three co-founders spent over a decade designing chips and had already built and sold one semiconductor company before starting Quanfluence, and LightSpeed Photonics' founders bring decades of experience in semiconductor packaging and long photonics-industry tenure, closely matching the original finding of a second-time entrepreneur with roughly a decade in photonics R&D. The other pattern is the academic-faculty spinout, exemplified by DeepLase (Dr. Deepak Jain, IIT Delhi faculty, with a team drawn directly from IIT Delhi's photonics research base). Engineering talent draws on IISc CeNSE's photonics group (which has worked with DRDO, ISRO and BEL) and, per DeepLase and Quanfluence, on IIT Delhi's and IIT Madras's photonics/IC-design research groups feeding company teams directly.

WHAT TO DO

Extend the CPPICS model from PIC research to PIC pilot fabrication.

MeitY already funds the CoE against India Semiconductor Mission objectives [318][317]; the gap is manufacturing, and ISM's ₹76,000 crore envelope is the instrument already in place.

Use TDF as a qualification pipeline, not just an R&D grant [330][331] — DRDO/ISRO co-development is already how photonics engineers are trained [328];

Composite 3.25 · 8 of 9 criteria · 52 companies tracked

Need  **4.0**
Readiness  **3.3**
Opportunity  **2.7**

Business model: IoT-sensor SaaS / subscription enterprise services

Two honest caveats, stated up front. (i) The NM-ICPS outlay value is not captured in the evidence record — the score is a token for "named in a funded national mission", nothing more [335]. [DATA GAP — needs the NM-ICPS Cabinet outlay figure].

SCORE BREAKDOWN — ALL 9 CRITERIA

STRATEGIC NEED

- **Sovereignty 4/5:** named in policy with budget line
- **Value-chain 4/5:** input dependency for ≥3 sectors
- **Import substitution:** data gap

READINESS

- **Talent 3/5:** 2k–10k / 4–7 institutions
- **Infrastructure 4/5:** 2–4 operational
- **Cluster deps. 3/5:** cluster exists with documented gaps

OPPORTUNITY

- **Global market 4/5:** \$100–500B
- **Domestic market 2/5:** \$500M–2B
- **Growth trajectory 2/5:** 6–12% CAGR

INFRASTRUCTURE & CLUSTER

The missing thing is the silicon layer under the box. Assembly clusters exist across seven states [340]; the gap is in inputs, and it is precisely the input gap that the missing import figure would have sized.

CAPITAL PATHWAY



Common entry point: a government scheme or mission, open to any qualifying company in the sector — not a private deal specific to the named company below.

● **Entry:** Working (public) — MeitY's TIDE scheme gives each empowered incubation centre grant-in-aid up to ₹155 lakh, of which ₹125 lakh is earmarked for incubating ICT/electronics companies [344]

Not enough evidence to establish a common investor pattern for bridge capital here — this is the only named transaction on record.

● **Bridge:** Working (corporate VC) — Altizon raised a \$4M Series A led by Wipro Ventures and Lumis Partners, with The Hive and Infuse Ventures [345]

Not enough evidence to establish a common investor pattern for growth capital here — this is the only named transaction on record.

● **Growth:** Working, but foreign and AI-framed — Qualcomm Ventures' Strategic AI Venture Fund commits up to \$150M to Indian startups [346]

Not enough evidence to call this a repeatable pattern — this is the only completed exit on record for the sector (India has completed exits in just 30 of 47 tracked sectors).

● **Exit:** One precedent — Zenatix Solutions was acquired by Hero Electronix in May 2018 and subsequently sold onward to a global buyer [347]

WHO IS BUILDING

52 companies tracked — the deepest base of the three, and the reason IoT reads as "already commercial" while scoring as "state must build". Stellapps (Bengaluru, Series C, \$50M), Altizon (Pune, Series A, \$13.8M), Zenatix (Gurugram, acquired, \$1.74M), plus Cavli Wireless (Kochi), A5G Networks (Pune), XYMA Analytics,...

SHIPS: **Stellapps** — SmartMoo, an IoT platform for milking systems, cattle wearables and cold-chain monitoring, used by dairy · **Altizon** — DFX, an AI-native manufacturing execution system turning shop-floor sensor data into quality/productivity · **Zenatix** — An IoT-powered building-automation and energy-monitoring platform for commercial buildings, deployed across 3,000+

TALENT SOURCING — COMMON PROPERTY CORPORATE-VETERAN TEAM EXITS

Among the five IoT candidates, a real pattern shows up in three companies: Stellapps, Altizon and Zenatix were each founded by multiple co-founders with 10–24 years of prior corporate experience — all five Stellapps founders came from Wipro, Altizon's trio came from enterprise-tech firms including Persistent Systems, Amazon, BMC Software and BladeLogic, and Zenatix's Rahul Bhalla and Vishal Bansal came from IIT Delhi-linked KPO and insurance-finance roles — teams of corporate veterans quitting together, not solo campus founders. That is a distinct pattern from the campus-team story already documented for Detect Technologies (shared with Industrial Software). Cavli Wireless and A5G Networks add no confirmable founder-background data. Engineering talent for IoT flows through corporate-alumni networks (Stellapps' Wipro alumni base, reinforced by incubation at IIT Madras's Rural Technology Business Incubator) and government-backed labs like Altizon's home base at Pune's C4i4 Labs Industry 4.0 initiative, rather than fresh campus hiring alone.

WHAT TO DO

Measure the import bill before funding substitution.

No IoT-specific import figure exists; the sector cannot make its sovereignty case without one [DATA GAP] . Attach the IoT vertical inside NM-ICPS to the manufacturing clusters, not only to campuses.

The TIH Foundation for IoT and IoE at IIT Bombay holds the vertical [335]; the ESDM base sits in seven other states [340]. Scale TIDE past incubation.

Composite 3.62 · 8 of 9 criteria · 5 companies tracked



Business model: B2B deployment of autonomous robots at scale

Need is carried by three 4s, each from a distinct rubric rule. The deficit is not infrastructure and not the cluster — both score 4. It is a thin academic base (6 institutions) feeding a sector whose commercially evidenced activity is not road autonomy at all.

SCORE BREAKDOWN — ALL 9 CRITERIA

STRATEGIC NEED

- **Sovereignty 4/5:** named in policy with budget line
- **Value-chain 4/5:** input dependency for ≥3 sectors
- **Import substitution:** data gap

READINESS

- **Talent 3/5:** 2k–10k / 4–7 institutions
- **Infrastructure 4/5:** 2–4 operational
- **Cluster deps. 4/5:** cluster needed and exists in India

OPPORTUNITY

- **Global market 3/5:** \$20–100B
- **Domestic market 3/5:** \$2–10B
- **Growth trajectory 4/5:** 20–30% CAGR

INFRASTRUCTURE & CLUSTER

The deficit is not infrastructure and not the cluster — both score 4. It is a thin academic base (6 tracked companies) relative to the ambition of full autonomy.

CAPITAL PATHWAY



Common entry point: private angel/VC money, not a government scheme — the named round below is a representative instance of a private-capital-first sector.

● **Entry:** Private venture/angel money, not government grants — General Autonomy (Bengaluru humanoid robotics) raised a ₹32 crore (~\$3.3M) seed led by existing investors including Elevation Capital [359]

Not enough evidence to establish a common investor pattern for bridge capital here — this is the only named transaction on record.

● **Bridge:** Foreign angel capital, with no domestic institutional VC or government grant involved — Swayatt Robots raised \$4M from a US-based investor [356]

Not enough evidence to establish a common investor pattern for growth capital here — this is the only named transaction on record.

● **Growth:** Breaks here. No verified Indian growth-stage round dedicated to a pure autonomous-systems (non-warehouse-robotics) company was found; the largest disclosed rounds sit in adjacent robotics/warehouse categories [356]

Not enough evidence to call this a repeatable pattern — this is the only completed exit on record for the sector (India has completed exits in just 30 of 47 tracked sectors).

● **Exit:** One evidenced route: acquisition by an Indian founder-led strategic building in-house autonomy — LAT Aerospace (co-founded by Deepinder Goyal and Surobhi Das) acquired a Gurgaon-based company [360]; target name truncated in the evidence record [DATA GAP]

WHO IS BUILDING

Swayatt Robots — Bhopal · \$7.1M **Minus Zero** — Bengaluru · Seed · \$1.7M **Flux Auto** — Bengaluru · Seed · \$1.2M

SHIPS: **Swayatt Robots** — An autonomous-driving software stack aimed at unstructured, chaotic Indian traffic conditions (no single named · **Minus Zero** — zPod, a vision-only, self-supervised full-stack AI platform for ADAS/autonomous driving licensed to OEMs · **Flux Auto** — Odin, a retrofit 'physical AI' kit that bolts onto existing forklifts, pallet jacks and tow trucks to automate

TALENT SOURCING — COMMON PROPERTY NO CORPORATE STINT — RESEARCH OR SERIAL-FOUNDER

The three Autonomous Systems companies with usable data share one trait: none of the founders came from a long corporate or national-lab career before founding. Swayatt Robots' Sanjeev Sharma returned from a research track (IIT Roorkee, a University of Alberta master's, declining a UMass Amherst PhD offer) straight into founding; Flux Auto's Pranav Manpuria and Abhishek Gupta had each already built and run an earlier startup of their own before founding Flux Auto; and Minus Zero's Gagandeep Reehal and Gursimran Kalra have no documented prior employer at all in available sources. blurgs.ai and Sirab Technologies Transportation had no findable information despite extensive search — a genuine data gap, not a guess. Engineering talent draws on academic labs building full self-driving stacks, most visibly IIIT Hyderabad (its own self-driving-car project and, per Minus Zero's investor list, its iHub-Data AI institute as a research/talent anchor).

WHAT TO DO

Publish the on-road position.

Before any instrument: an unambiguous, citable statement of what is and is not permitted for on-road autonomous testing and deployment.

If the answer is restrictive, a funding response is the wrong response, and every rupee of grant capital into road autonomy is misallocated.

Composite unranked · 6 of 9 criteria · 19 companies tracked



Business model: B2B inspection services + emerging defence hardware

Note the cap rule: the mission's outlay was large enough to fire the top sovereignty rule but small enough to be capped back to 4. Note also that three of nine criteria are unscored, which is why no composite is computed for this sector (6 of 9 criteria scored).

SCORE BREAKDOWN — ALL 9 CRITERIA

STRATEGIC NEED	READINESS	OPPORTUNITY
● Sovereignty 4/5: named in policy with budget line ● Value-chain 4/5: input dependency for ≥3 sectors ⦿ Import substitution: data gap	● Talent 4/5: 10k–50k specialists / 8–14 institutions ● Infrastructure 4/5: 2–4 operational ● Cluster deps. 3/5: cluster exists with documented gaps	● Global market 2/5: \$5–20B ⦿ Domestic market: data gap ⦿ Growth trajectory: data gap

INFRASTRUCTURE & CLUSTER

The specific missing thing is the fabrication yard, not the lab. Talent scores 4 and national facilities score 4. What scores 3 is the industrial base a subsea hardware company must buy from: an existing but incompletely served shipbuilding and offshore-fabrication cluster [366].

CAPITAL PATHWAY

●

Entry

●

Bridge

●

Growth

⦿

Exit

WHO IS BUILDING

19 companies tracked — by far the largest base of the three, and geographically distinctive: Kochi and Chennai, not Bengaluru.

EyeROV (IROV Technologies Pvt Ltd) — Kochi · Seed · \$3.4M

Planys Technologies — Chennai · Growth

PierSight — Ahmedabad

TALENT SOURCING — COMMON PROPERTY IIT STUDENT-TEAM PIPELINE

Common thread: IIT student-competition teams moving directly into founding. Both companies trace to IIT Delhi/Madras underwater-robotics competition teams, with a single faculty co-founder providing the only senior-veteran presence.

WHAT TO DO

Open a startup-accessible window inside the Deep Ocean Mission.

The instrument exists and is funded at ₹4,077 crore over five years [361]; the gap is standing, not size.

Today's evidenced founder-facing instrument is iDEX/SPRINT [369] — a defence channel, which constrains every non-defence ocean application to a defence use case.

Proof It Works: Six Companies to Know

The priority calls above rest on company-level evidence, not just scores. Six names spanning AI, space, semiconductors and an exit — as concrete reference points.

Sarvam AI

Foundation Models · Bengaluru

Raised \$234M of a \$300M Series B at a \$1.5B valuation (2026) — India's newest AI unicorn, largest DeepTech round of the year.

Krutrim

Foundation Models · Bengaluru

Hit a \$1B valuation in Jan 2024 — the fastest Indian company on record to reach unicorn status.

Skyroot Aerospace

SpaceTech · Hyderabad

\$1.1B pre-money valuation round made it India's first space-tech unicorn, three years after launching the country's first private suborbital rocket.

Fractal Analytics

Artificial Intelligence · Mumbai

Completed India's first AI-focused IPO (Feb 2026, ₹2,833 cr issue) — 26 years after founding in 2000.

Uniphore

Generative AI · Chennai

An IIT Madras incubatee in 2008 reached \$870M total funding and a \$2.5B valuation, with NVIDIA and AMD on the cap table.

Steradian Semiconductors

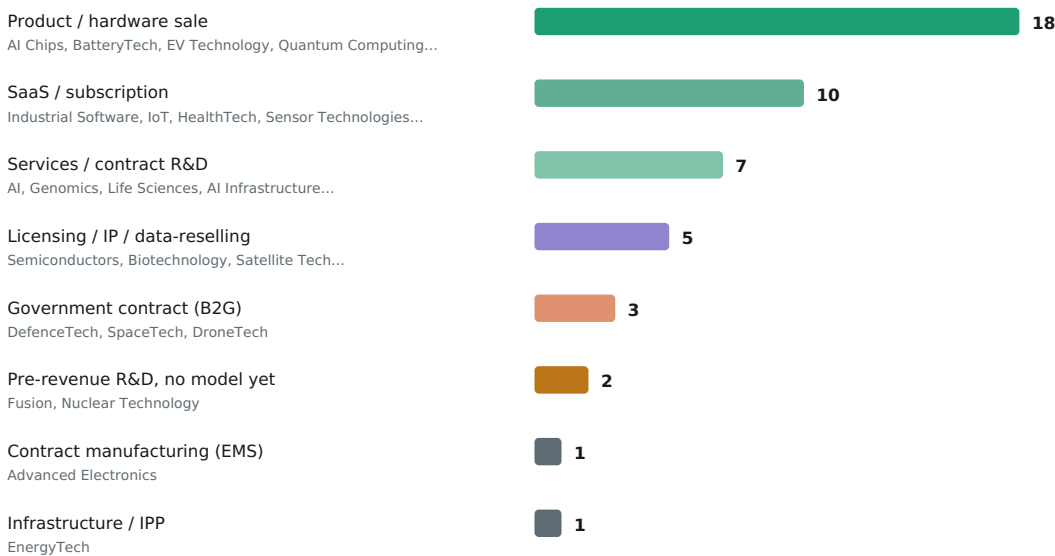
Semiconductors · Bengaluru

A fabless mmWave radar-chip startup acquired by Renesas Electronics (2022) — a rare Indian semiconductor IP exit to a global tier-one.

How These Companies Actually Work

What a company in each sector actually looks like: founder and engineering talent sources, business model. Talent findings are exemplars from one company per sector, not distributions — read as evidence that a path exists, not a measurement of how common it is.

Business model — all 47 sectors classified



Founder and engineering talent — common feeder pathways

Founder talent — common feeder pathways

A pattern list, not a distribution — each finding is drawn from one exemplar company, not measured across a sector's workforce

- **IIT / IISc academic origin**
e.g. AI Chips, Artificial Intelligence, Advanced Electronics
- **Corporate veteran / spinout**
e.g. AI Chips, Advanced Manufacturing, AgriTech DeepTech
- **DRDO / ISRO / defence-PSU**
e.g. Digital Twin, IoT, Material Science
- **Returnee / diaspora PhD**
e.g. AI Infrastructure, AR/VR/XR, Autonomous Systems
- **Serial / repeat founder**
e.g. AI Infrastructure, Biotechnology, Cybersecurity

Patterns are not mutually exclusive and not ranked by prevalence — a sector can show more than one, and this list does not imply which is most common.

Engineering talent — common feeder pathways

A pattern list, not a distribution — each finding is drawn from one exemplar company, not measured across a sector's workforce

- **IIT / IISc / IIIT / NIT pipeline**
e.g. AI Chips, AgriTech DeepTech, ClimateTech
- **Academic research lab (non-IIT)**
e.g. AR/VR/XR, Foundation Models, Hydrogen
- **MNC captive R&D / GCC centres**
e.g. AI Infrastructure, Advanced Electronics, Cybersecurity
- **Same-sector industry veteran**
e.g. BatteryTech, Biotechnology, EnergyTech
- **In-house / founder-built**
e.g. AgriTech DeepTech, DroneTech

Patterns are not mutually exclusive and not ranked by prevalence — a sector can show more than one, and this list does not imply which is most common.

Where It's Happening: The City Map

India's deep-tech geography is sharply concentrated. Bengaluru, Hyderabad, Delhi NCR, Mumbai, Chennai and Pune capture the overwhelming majority of research anchors, venture capital, corporate R&D and specialist talent — while 26 emerging tier-2/tier-3 cities already host 7,000+ startups and drew 13% of 2022 startup funding, the decentralisation counter-trend.

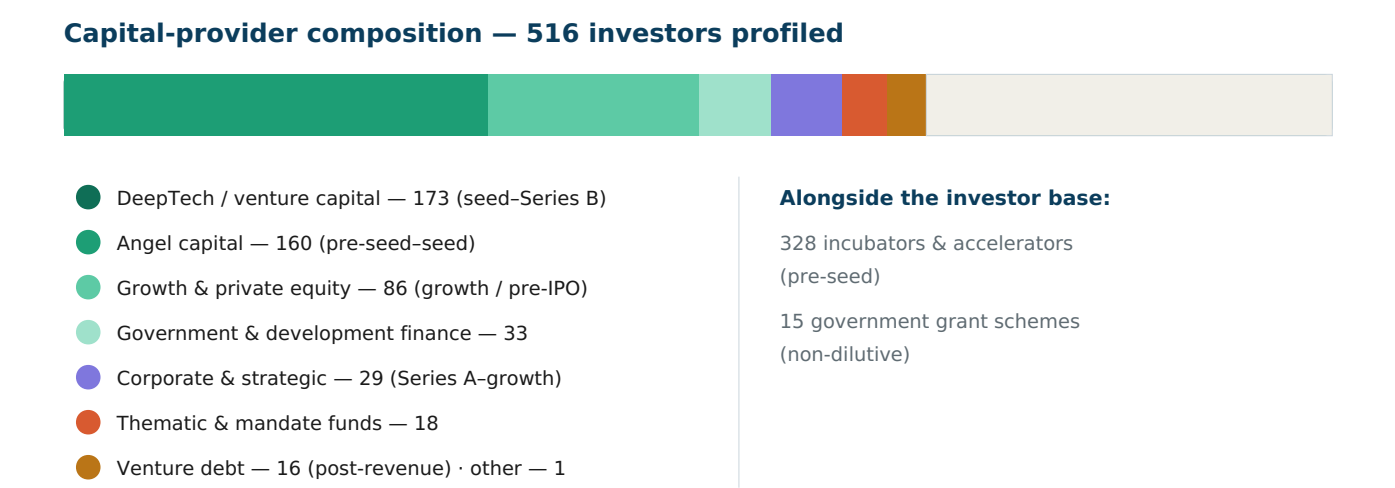
City	What the evidence shows
Bengaluru	58% of India's AI startup funding; 900+ GCC units (34–39% of India's GCC market); ~\$79B cumulative VC since 2010 (46% of national total)
Hyderabad	494 deep-tech companies tracked, 141 funded, 34 at Series A+; \$742M deep-tech raised; 515+ GCCs, healthcare-biotech leader
Delhi NCR	~1,200 deep-tech companies, 276 funded, 64 at Series A+; \$1.53B deep-tech raised; ~270 GCCs (15–18% of India)
Mumbai	500 deep-tech companies, 131 funded, 40 at Series A+; \$1.25B deep-tech raised; BFSI is the primary GCC sector of excellence
Pune	1,214 startups (~17/100k), 4 unicorns; \$4.7B all-startup funding over a decade; 500+ GCCs, 340,000 GCC employees
Chennai	966 startups, 1 unicorn; \$2.68B all-startup funding over a decade; the IIT Madras incubator (IITMIC) alone backs a 567-startup portfolio

Coverage caveat: a headquarters city is mapped for only a minority of the 1,400 tracked companies. Figures blend one primary dataset with third-party trackers (Tracxn, StartupBlink) labelled at source.

The Capital & Exit Map

How capital reaches each sector, rung by rung (entry, bridge to Series A, growth, exit), and who is actually providing and receiving it.

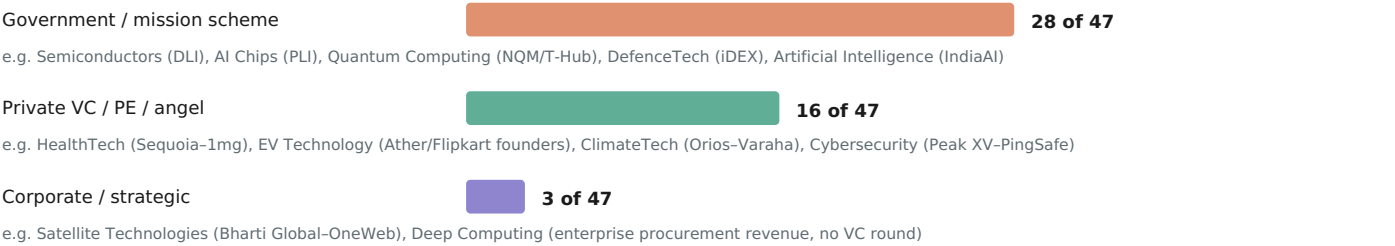
Capital-provider composition (516 investors)



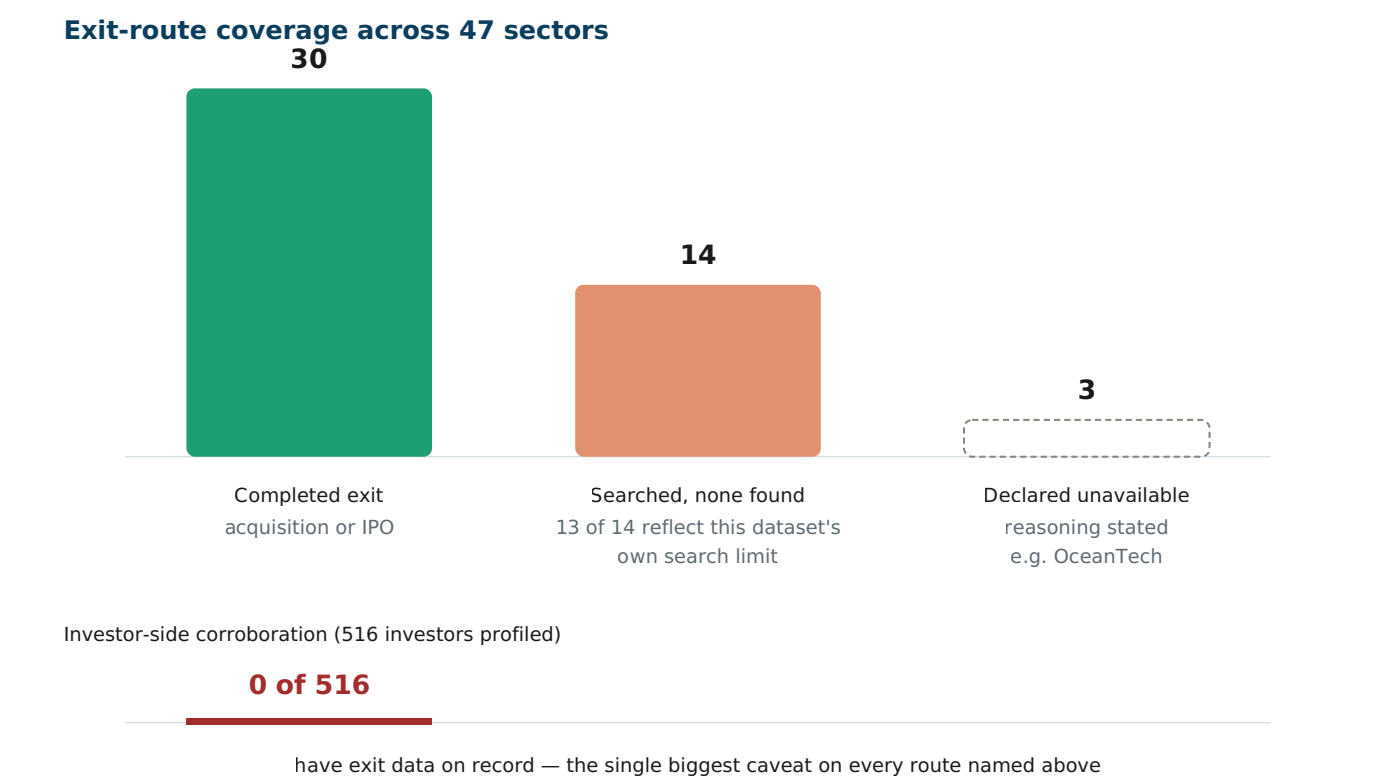
Source: counted from the 516 investors profiled here

Entry-capital typology — how the first cheque actually arrives

Entry-capital typology — all 47 sectors



Exit reality



The State's Hand: Missions and Schemes Behind the Numbers

India's deep-tech push is financed less by one line item than by a portfolio of mission-mode programmes across DST, MeitY, Defence, Space and DPIIT. These are the schemes that recur across the sector dossiers' sovereignty-value and entry-capital findings — six of the largest, by headline size.

Scheme	Agency	Outlay	USD	Why it matters
Semiconductor Mission (Semicon 2.0)	DST/ MeitY	₹1,27,500 cr	~\$15.4B	Largest single commitment; underwrites the sovereignty case for Semiconductors.
DST RDI Scheme	DST	₹1,00,000 cr / 6 yrs	~\$12B	Broad research-and-innovation financing across ministries.
IndiaAI Mission	MeitY	₹10,371.92 cr	~\$1.25B	Funds foundation-model consortia (Sarvam, BharatGen and others).
National Quantum Mission	DST	₹6,003.65 cr / 8 yrs	~\$723M	Launched Apr 2023; anchors Quantum Computing's sovereignty score.
SIDBI Fund of Funds	SIDBI	₹10,000 cr	~\$1.2B	Recycles public capital into private VC funds.
Startup India Seed Fund	DPIIT	₹945 cr	~\$114M	Early-stage, non-dilutive seed capital.

The structural read: capital is no longer the binding constraint at the mission level. Gaps remain in a few agency-specific outlays not separately disclosed, and in cross-source reconciliation between trackers.